
ASSIGNMENT-1

Name: Jeevan P
Batch: G4-25VLSI

The domain of the Project:
Physical Design (VLSI)

Linux Command

chmod – Change File Permissions

The chmod command is used to change the access permissions of a file or directory in Linux. It controls who can read, write, or execute a file. The permission is represented by three digits, where the first digit is for the owner, the second for the group, and the third for others. Each digit is a combination of read (4), write (2), and execute (1).

- It defines access control for files and directories
- Uses numeric values (4 = read, 2 = write, 1 = execute)
- Three digits represent owner, group, and others

Example:

```
chmod 642 file.txt
```

Explanation:

- Owner → read + write (6)
- Group → read (4)
- Others → write (2)

grep – Search Pattern in File

The grep command is used to search for specific patterns or words inside a file. It processes the file line by line and prints the entire line wherever the pattern is found. It is very useful for filtering logs and large text files.

- Searches text inside files
- Works line by line
- Returns matching lines

Example:

```
grep -i "error" log.txt
```

Explanation:

- -i ignores case sensitivity
- It will match "Error", "ERROR", etc.
- Displays all lines containing the word "error"

awk – Text Processing Tool

The awk command is used for processing and analyzing structured text data. It reads files line by line, where each line is called a record, and each word is called a field. It is commonly used to extract specific columns from data files.

- Processes text data field-wise
- Each line = record
- Each word = field

Example:

```
awk '{print $1, $2}' data.txt
```

Explanation:

- \$1 → first field
- \$2 → second field
- Prints first and second columns from the file

vi – Text Editor

The vi command is used to open and edit text files in Linux. It operates in different modes such as command mode, insert mode, and last line mode. It is widely used for editing configuration files.

- Used to create and edit files
- Has different modes (command, insert, last line)
- Efficient for terminal-based editing

Example:

```
vi file.txt
```

Explanation:

- Opens the file in editor
- Press i to enter insert mode
- Press Esc then type :wq to save and exit

Basic Commands

- **ls – List Files**

The ls command is used to display files and directories in the current location.

- Shows directory contents
- Can show hidden files and details

Example:

```
ls -l
```

Explanation:

- -l shows detailed information like permissions, size, and date

- **cd – Change Directory**

The cd command is used to move between directories in the system.

- Helps in navigation
- Can move forward or backward

Example:

```
cd ..
```

Explanation:

- Moves one directory back

- **pwd – Print Working Directory**

The pwd command shows the current directory path where the user is working.

- Displays full path
- Useful for navigation

Example:

```
pwd
```


Explanation:

- Outputs current directory location

- **mkdir – Create Directory**

The mkdir command is used to create a new directory.

- Creates folders
- Helps organize files

Example:

```
mkdir project
```

Explanation:

- Creates a folder named "project"

- **rm – Remove Files**

The rm command is used to delete files or directories.

- Deletes files permanently
- Can remove folders using options

Example:

```
rm file.txt
```

Explanation:

- Deletes the file named "file.txt"

OpenLane Environment

OpenLane is an automated RTL-to-GDSII flow used for digital ASIC design. It works on top of tools like OpenROAD, Magic, and Netgen, and is typically run inside a Docker-based environment or local Linux setup. The environment includes all required EDA tools and configuration files needed for chip design.

- Provides complete ASIC design flow (synthesis → layout → verification)
- Uses pre-installed open-source tools
- Runs using Docker or local installation
- Requires PDK (Process Design Kit) like Sky130

Installation of Git

Git is a distributed version control system used to manage and track changes in source code. It is mainly used to clone repositories from platforms like GitHub, which is necessary for downloading the OpenLane project. Installing Git ensures that the user can access, update, and manage project files efficiently.

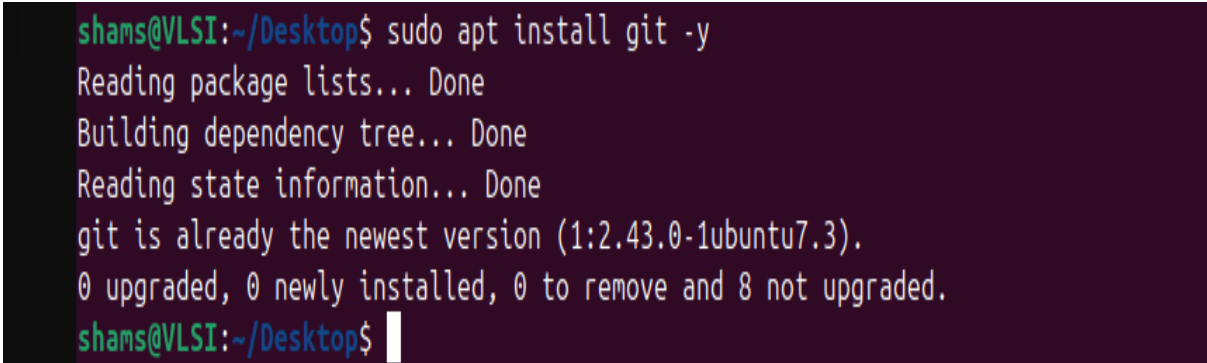
- Used for cloning repositories (e.g., OpenLane from GitHub)
- Helps in version control and tracking changes
- Lightweight and widely used in development environments

Example:

```
sudo apt install git -y
```

Explanation:

- `sudo apt install git -y` → Installs Git without manual confirmation



```
shams@VLSI:~/Desktop$ sudo apt install git -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
git is already the newest version (1:2.43.0-1ubuntu7.3).
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
shams@VLSI:~/Desktop$
```

Installation of Docker

Docker is a containerization platform that allows applications like OpenLane to run in an isolated environment with all dependencies included. It ensures consistency across different systems and avoids compatibility issues. Docker is essential for running OpenLane smoothly.

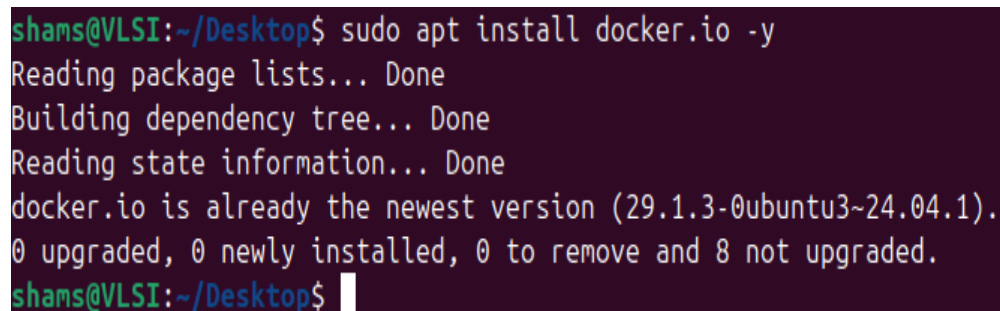
- Provides isolated environment using containers
- Avoids dependency conflicts
- Required to run OpenLane flow

Example:

```
sudo apt install docker.io -y
```

Explanation:

- `sudo apt install docker.io -y` → Installs Docker engine

A terminal window with a dark purple background. The prompt is 'shams@VLSI:~/Desktop\$'. The command 'sudo apt install docker.io -y' is entered. The output shows 'Reading package lists... Done', 'Building dependency tree... Done', and 'Reading state information... Done'. It then states 'docker.io is already the newest version (29.1.3-0ubuntu3~24.04.1). 0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.' The prompt returns to 'shams@VLSI:~/Desktop\$' with a cursor.

```
shams@VLSI:~/Desktop$ sudo apt install docker.io -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (29.1.3-0ubuntu3~24.04.1).
0 upgraded, 0 newly installed, 0 to remove and 8 not upgraded.
shams@VLSI:~/Desktop$
```

Post Installation Setup (Docker)

After installing Docker, some additional setup steps are required to start and configure the service.

Example:

```
sudo systemctl start docker
sudo systemctl enable docker
sudo usermod -aG docker $USER
```

Explanation:

- `systemctl start docker` → Starts Docker service
- `systemctl enable docker` → Enables auto-start at boot
- `usermod -aG docker $USER` → Allows user to run Docker without sudo

```
shams@VLSI:~/Desktop$ git clone https://github.com/The-OpenROAD-Project/OpenLane
git
fatal: destination path 'OpenLane' already exists and is not an empty directory.
shams@VLSI:~/Desktop$ cd OpenLane
shams@VLSI:~/Desktop/OpenLane$ ls
AUTHORS.md      designs      gui.py       Makefile     scripts
configuration  docker      install     README.md   tests
CONTRIBUTING.md docs        Jenkinsfile regression_results venv
default.cvcrc  env.py     klayoutrc   requirements.txt
dependencies    flow.tcl   LICENSE     run_designs.py
shams@VLSI:~/Desktop/OpenLane$
```

Terminal

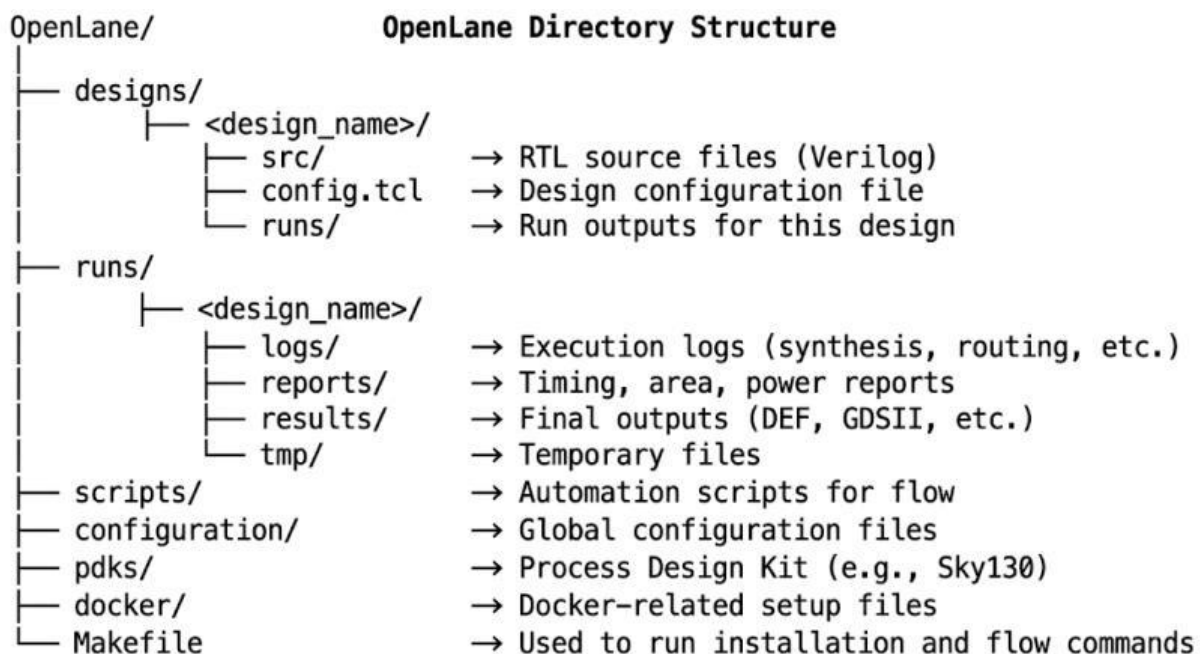


Fig: - Directory Structure of OpenLane

Reports in OpenLane

OpenLane generates multiple reports at different stages of the design flow. These reports help analyze performance, timing, area, and power of the design.

- Located inside runs/<design_name>/reports/
- Includes synthesis, floorplan, placement, routing reports
- Used for debugging and optimization

Example:

```
cd runs/spm/reports/synthesis
ls
```

Explanation:

- Opens synthesis report folder
- Displays files like area report, timing report

Analysis of Logs

Logs are generated during each step of the OpenLane flow and are useful for debugging errors and understanding tool execution.

- Located inside runs/<design_name>/logs/
- Separate logs for synthesis, placement, routing
- Helps identify errors and warnings

Example:

```
cd runs/spm/logs/synthesis
less synthesis.log
```

Explanation:

- Opens synthesis log file
- Shows step-by-step execution details

TCL Scripting

TCL (Tool Command Language) scripting is a simple and powerful scripting language widely used in VLSI design automation tools, including OpenLane, for controlling and automating the ASIC design flow. It is primarily used to define configurations, execute commands, and manage different stages such as synthesis, floorplanning, placement, and routing. In OpenLane, TCL scripts (like config.tcl) specify design parameters such as clock period, design name, input files, and constraints, allowing users to customize the flow without modifying the core tool. TCL scripts are easy to write and follow a command-based structure where each line represents an instruction executed sequentially. This scripting capability helps automate repetitive tasks, ensures consistency in design execution, and enables efficient control over the entire chip design process.

Example:

```

set ::env(DSIGN_NAME) "spm"
set ::env(CLOCK_PERIOD) 10
set ::env(VERILOG_FILES) "designs/spm/src/spm.v"
set ::env(SYNTH_STRATEGY) "AREA 0"

```

TCL scripting in OpenLane acts as the **backbone of automation**, enabling users to control the entire chip design flow efficiently through simple commands.

```

shams@VLSI:~/Desktop/OpenLane$ make mount
cd /home/shams/Desktop/OpenLane && \
    docker run --rm -v /home/shams/Desktop/OpenLane:/openlane -v /home/shams/
/Desktop/OpenLane/designs:/openlane/install -v /home/shams:/home/shams -v /home/
shams/.ciel:/home/shams/.ciel -e PDK_ROOT=/home/shams/.ciel -e PDK=sky130A --us
er 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --sec
urity-opt seccomp=unconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65
b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$

```

Fig: - Moving into Docker using Command to run the design

```

OpenLane Container (ff5509f):/openlane$ ./flow.tcl -design spm
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/spm/config.json'...
[INFO]: PDK Root: /home/shams/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/spm/runs/RUN_2026.05.03_19.28.09
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
[INFO]: Running linter (Verilator) (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs/sy
[INFO]: 0 errors found by linter
[WARNING]: 1 warnings found by linter
[STEP 1]
[INFO]: Running Synthesis (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs/synthesis/1
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/spm/runs/RUN_2026.05.0
.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs
[WARNING]: Current core area is too small for the power grid settings chosen. The power g
[INFO]: Floorplanned with width 90.62 and height 89.76.
[STEP 4]
[INFO]: Running IO Placement (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs/floorpla
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs/f
[INFO]: Power planning with power {VPWR} and ground {VGND}...

```



```

[INFO]: Running LVS (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs/signoff/39-lvs.le
[STEP 40]
[INFO]: Running Magic DRC (log: designs/spm/runs/RUN_2026.05.03_19.28.09/logs/signoff/40-
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/spm/runs/RUN_2026.05.03_19.28
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/spm/runs/RUN_2026.05.03_19.28.
[INFO]: Saving current set of views in 'designs/spm/runs/RUN_2026.05.03_19.28.09/results/
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/spm/runs/RUN_2026.05.03_19.28.09/rep
[Trash] Created metrics report at 'designs/spm/runs/RUN_2026.05.03_19.28.09/reports/metri
[WARNING]: There are max fanout violations in the design at the typical corner. Please re
26.05.03_19.28.09/reports/signoff/31-rcx_sta.checks.rpt'.
[INFO]: There are no hold violations in the design at the typical corner.
[INFO]: There are no setup violations in the design at the typical corner.
[SUCCESS]: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: 1 warnings found by linter
[WARNING]: Current core area is too small for the power grid settings chosen. The power g
[WARNING]: Module sky130_fd_sc_hd__tapvpwrgnd_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values m
[WARNING]: There are max fanout violations in the design at the typical corner. Please re
26.05.03_19.28.09/reports/signoff/31-rcx_sta.checks.rpt'.

```

Fig: - Showing a run of a sample design flow

```

OpenLane Container (ff5509f):/openlane$ cd designs
OpenLane Container (ff5509f):/openlane/designs$ ls
ci  spm
OpenLane Container (ff5509f):/openlane/designs$ cd spm
OpenLane Container (ff5509f):/openlane/designs/spm$ ls
config.json  pin_order.cfg  runs  src
OpenLane Container (ff5509f):/openlane/designs/spm$ cd runs
OpenLane Container (ff5509f):/openlane/designs/spm/runs$ ls
openlane_test      RUN_2026.04.30_16.18.57  RUN_2026.04.30_16.24.18
RUN_2026.04.30_16.17.31  RUN_2026.04.30_16.20.05  RUN_2026.05.03_19.28.09
OpenLane Container (ff5509f):/openlane/designs/spm/runs$

```

Fig: - Output Result in the RUN destination File

The successful completion of the **OpenLane** run indicates that the entire digital design flow—from synthesis through floorplanning, placement, routing, and signoff checks—has executed correctly, producing a valid GDSII layout from the RTL description. Although several warnings were reported, they are typical for a small test design like *spm* and do not indicate critical failures. The max fanout violation suggests that a signal is driving more gates than recommended, while the core area warning

reflects a compact layout that required automatic adjustment of the power grid. The presence of blackboxed standard cells simply means certain library elements were not fully expanded during timing

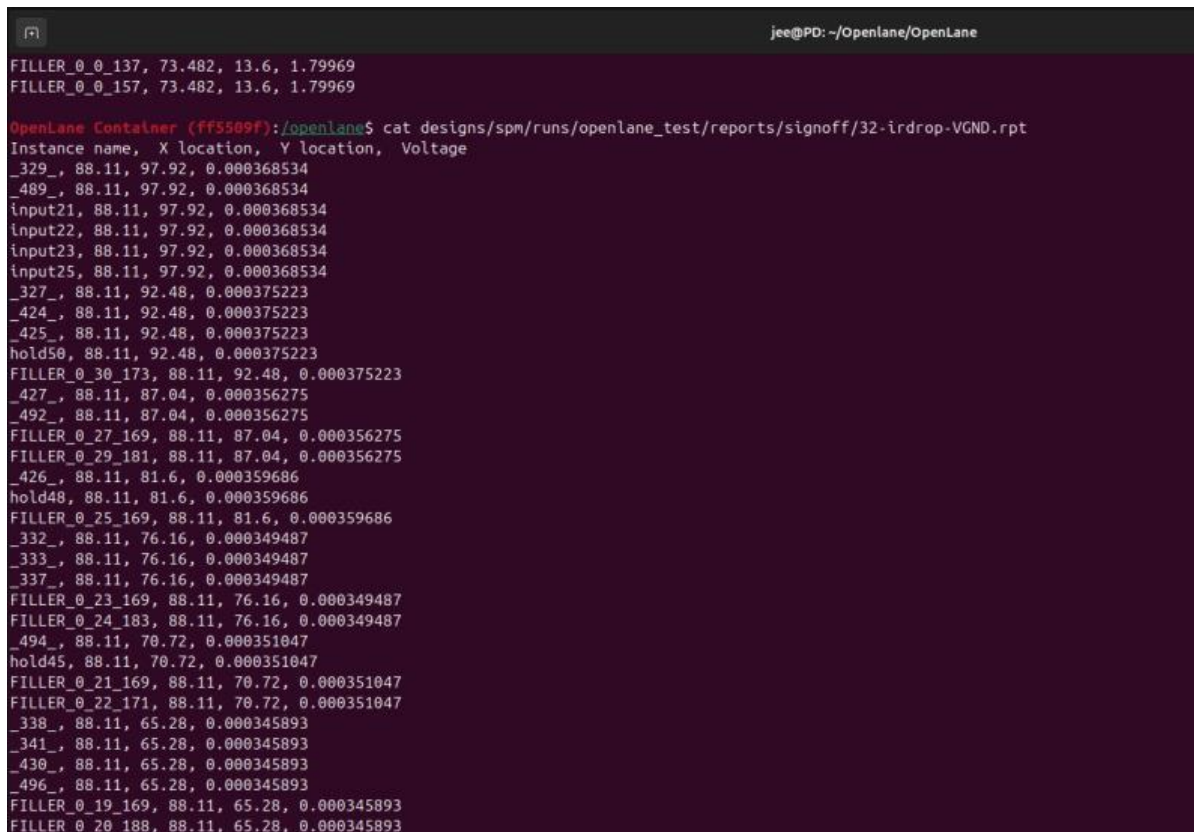
analysis, which is normal in this context. Additionally, the absence of VSRC location files implies that IR drop analysis results may be approximate. Overall, despite these minor warnings, the flow completed successfully and the generated layout is valid for analysis and learning purposes.

Assignment 1 – TCL Automation, Wire Length and IR Drop Analysis

This section includes TCL automation tasks performed in OpenLane for wire length calculation, clock wire length analysis, data wire length extraction, STA report analysis, and IR drop analysis.

Tasks Completed:

1. Total wire length calculation using TCL scripting
2. Clock wire length extraction
3. Data wire length extraction
4. IR drop analysis for VDD and VSS
5. Timing report and slack analysis
6. Routing report exploration

A terminal window with a dark background and light text. The prompt is 'jee@PD: ~/Openlane/OpenLane'. The command executed is 'cat designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VGND.rpt'. The output is a list of instance names, X and Y coordinates, and voltage levels for various components in the design.

```
jee@PD: ~/Openlane/OpenLane
FILLER_0_0_137, 73.482, 13.6, 1.79969
FILLER_0_0_157, 73.482, 13.6, 1.79969

OpenLane Container (ff5509f):/openlane$ cat designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VGND.rpt
Instance name, X location, Y location, Voltage
_329_, 88.11, 97.92, 0.000368534
_489_, 88.11, 97.92, 0.000368534
input21, 88.11, 97.92, 0.000368534
input22, 88.11, 97.92, 0.000368534
input23, 88.11, 97.92, 0.000368534
input25, 88.11, 97.92, 0.000368534
_327_, 88.11, 92.48, 0.000375223
_424_, 88.11, 92.48, 0.000375223
_425_, 88.11, 92.48, 0.000375223
hold50, 88.11, 92.48, 0.000375223
FILLER_0_30_173, 88.11, 92.48, 0.000375223
_427_, 88.11, 87.04, 0.000356275
_492_, 88.11, 87.04, 0.000356275
FILLER_0_27_169, 88.11, 87.04, 0.000356275
FILLER_0_29_181, 88.11, 87.04, 0.000356275
_426_, 88.11, 81.6, 0.000359686
hold48, 88.11, 81.6, 0.000359686
FILLER_0_25_169, 88.11, 81.6, 0.000359686
_332_, 88.11, 76.16, 0.000349487
_333_, 88.11, 76.16, 0.000349487
_337_, 88.11, 76.16, 0.000349487
FILLER_0_23_169, 88.11, 76.16, 0.000349487
FILLER_0_24_183, 88.11, 76.16, 0.000349487
_494_, 88.11, 70.72, 0.000351047
hold45, 88.11, 70.72, 0.000351047
FILLER_0_21_169, 88.11, 70.72, 0.000351047
FILLER_0_22_171, 88.11, 70.72, 0.000351047
_338_, 88.11, 65.28, 0.000345893
_341_, 88.11, 65.28, 0.000345893
_430_, 88.11, 65.28, 0.000345893
_496_, 88.11, 65.28, 0.000345893
FILLER_0_19_169, 88.11, 65.28, 0.000345893
FILLER_0_20_188, 88.11, 65.28, 0.000345893
```

Fig 1: IR Drop Report for VGND

```

jee@PD: ~/Openlane/OpenLane

10 0.05      0.10 0.00 2.28 v _440_/A (sky130_fd_sc_hd__inv_2)
      0.04 0.07 2.35 ^ _440_/Y (sky130_fd_sc_hd__inv_2)
1 0.00      0.04 0.00 2.35 ^ _094_ (net)
      2.35 ^ _504_/RESET_B (sky130_fd_sc_hd__dfrtp_1)
      2.35 data arrival time

      10.00 10.00 clock clk (rise edge)
      0.00 10.00 clock source latency
      0.07 0.05 10.05 ^ clk (in)
1 0.01      0.07 0.00 10.05 ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
      0.10 0.18 10.22 ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
8 0.08      0.10 0.00 10.22 ^ clkbuf_3_5_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
      0.05 0.14 10.37 ^ clkbuf_3_5_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
9 0.03      0.05 0.00 10.37 ^ clknet_3_5_leaf_clk (net)
      -0.25 10.12 ^ _504_/CLK (sky130_fd_sc_hd__dfrtp_1)
      0.00 10.12 clock uncertainty
      0.23 10.12 clock reconvergence pessimism
      10.35 library recovery time
      10.35 data required time

-----
      10.35 data required time
      -2.35 data arrival time

-----
      8.00 slack (MET)

OpenLane Container (ff5509f):/openlane$ ^C
OpenLane Container (ff5509f):/openlane$ tclsh designs/spm/wire_length.tcl
couldn't read file "designs/spm/wire_length.tcl": no such file or directory
OpenLane Container (ff5509f):/openlane$ nano designs/spm/wire_length.tcl
bash: nano: command not found
OpenLane Container (ff5509f):/openlane$ tclsh designs/spm/wire_length.tcl
couldn't read file "designs/spm/wire_length.tcl": no such file or directory
OpenLane Container (ff5509f):/openlane$

```

Fig 2: STA Slack Report and TCL Execution

```

jee@PD: ~/Openlane/OpenLane

OpenLane Container (ff5509f):/openlane$ ^C
OpenLane Container (ff5509f):/openlane$ tclsh designs/spm/wire_length.tcl
couldn't read file "designs/spm/wire_length.tcl": no such file or directory
OpenLane Container (ff5509f):/openlane$ nano designs/spm/wire_length.tcl
bash: nano: command not found
OpenLane Container (ff5509f):/openlane$ tclsh designs/spm/wire_length.tcl
couldn't read file "designs/spm/wire_length.tcl": no such file or directory
OpenLane Container (ff5509f):/openlane$ cat > designs/spm/wire_length.tcl << 'EOF'
> set file [open "designs/spm/runs/openlane_test/reports/routing/24-wire_lengths.csv" r]
> set total 0
> set clock 0
> set data 0
> gets $file
> while {[gets $file line] >= 0} {
>     set parts [split $line ","]
>     set net_name [lindex $parts 0]
>     set length [lindex $parts 1]
>     set total [expr {$total + $length}]
>     if {[string match "*clk*" $net_name]} {
>         set clock [expr {$clock + $length}]
>     } else {
>         set data [expr {$data + $length}]
>     }
> }
> close $file
> puts "======"
> puts "Total Wire Length : $total um"
> puts "Clock Wire Length : $clock um"
> puts "Data Wire Length : $data um"
> puts "======"
> EOF
OpenLane Container (ff5509f):/openlane$ tclsh designs/spm/wire_length.tcl
======"
Total Wire Length : 6716.815000000004 um
Clock Wire Length : 784.48 um
Data Wire Length : 5932.335 um
======"
OpenLane Container (ff5509f):/openlane$

```

Fig 3: TCL Script for Wire Length Calculation

```
jee@PD: ~/Openlane/OpenLane

Clock Wire Length : 784.48 um
Data Wire Length : 5932.335 um
=====
OpenLane Container (ff5509f):/openlane$ cat designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VPWR.rpt
Instance name, X location, Y location, Voltage
_329_, 84.81, 100.64, 1.79975
input20, 84.81, 100.64, 1.79975
FILLER_0_32_165, 84.81, 100.64, 1.79975
_425_, 84.81, 95.2, 1.79972
_489_, 84.81, 95.2, 1.79972
input21, 84.81, 95.2, 1.79972
input22, 84.81, 95.2, 1.79972
input23, 84.81, 95.2, 1.79972
FILLER_0_31_165, 84.81, 95.2, 1.79972
_327_, 84.81, 89.76, 1.79971
_424_, 84.81, 89.76, 1.79971
_492_, 84.81, 89.76, 1.79971
FILLER_0_28_161, 84.81, 89.76, 1.79971
FILLER_0_29_181, 84.81, 89.76, 1.79971
FILLER_0_30_173, 84.81, 89.76, 1.79971
_427_, 84.81, 84.32, 1.79973
_491_, 84.81, 84.32, 1.79973
FILLER_0_27_169, 84.81, 84.32, 1.79973
_333_, 84.81, 78.88, 1.79974
_426_, 84.81, 78.88, 1.79974
FILLER_0_25_169, 84.81, 78.88, 1.79974
_337_, 84.81, 73.44, 1.79977
_340_, 84.81, 73.44, 1.79977
FILLER_0_22_171, 84.81, 73.44, 1.79977
FILLER_0_23_169, 84.81, 73.44, 1.79977
FILLER_0_24_183, 84.81, 73.44, 1.79977
_338_, 84.81, 68, 1.79976
_341_, 84.81, 68, 1.79976
_494_, 84.81, 68, 1.79976
hold45, 84.81, 68, 1.79976
FILLER_0_20_165, 84.81, 68, 1.79976
FILLER_0_21_165, 84.81, 68, 1.79976
FILLER_0_21_169, 84.81, 68, 1.79976
```

Fig 4: IR Drop Report for VPWR

```
jee@PD: ~/Openlane/OpenLane

FILLER_0_0_151, 76.782, 10.88, 0.000253671
FILLER_0_0_165, 76.782, 10.88, 0.000253671
FILLER_0_1_161, 76.782, 10.88, 0.000253671

OpenLane Container (ff5509f):/openlane$ cat designs/spm/runs/openlane_test/reports/signoff/31-rcx_sta.max.rpt

=====
report_checks -path_delay max (Setup)
=====
===== Typical Corner =====

Startpoint: _444_ (rising edge-triggered flip-flop clocked by clk)
Endpoint: p (output port clocked by clk)
Path Group: clk
Path Type: max

Fanout    Cap    Slew    Delay    Time    Description
-----
          0.00    0.00    0.00    0.00    clock clk (rise edge)
          0.00    0.00    0.00    0.00    clock source latency
          0.07    0.05    0.05    0.05    ^ clk (in)
          1      0.01    0.07    0.00    0.05    ^ clk (net)
          8      0.08    0.10    0.00    0.25    ^ clkbuf_0_clk/A (sky130_fd_sc_hd__clkbuf_16)
          8      0.08    0.10    0.19    0.25    ^ clkbuf_0_clk/X (sky130_fd_sc_hd__clkbuf_16)
          9      0.03    0.10    0.00    0.25    ^ clkbuf_3_0_f_clk/A (sky130_fd_sc_hd__clkbuf_16)
          9      0.03    0.05    0.16    0.41    ^ clkbuf_3_0_f_clk/X (sky130_fd_sc_hd__clkbuf_16)
          1      0.00    0.05    0.00    0.41    ^ clknet_3_0_leaf_clk (net)
          1      0.00    0.04    0.37    0.78    v _444_/CLK (sky130_fd_sc_hd__dftrtp_1)
          1      0.00    0.04    0.00    0.78    v _444_/Q (sky130_fd_sc_hd__dftrtp_1)
          1      0.03    0.08    0.19    0.96    v net35 (net)
          1      0.03    0.08    0.00    0.96    v output35/A (sky130_fd_sc_hd__clkbuf_4)
          1      0.03    0.08    0.19    0.96    v output35/X (sky130_fd_sc_hd__clkbuf_4)
          1      0.03    0.08    0.00    0.96    v p (net)
          1      0.03    0.08    0.00    0.96    v p (out)
          1      0.03    0.08    0.00    0.96    data arrival time
          10.00   10.00   10.00   10.00    clock clk (rise edge)
          10.00   10.00   10.00   10.00    clock network delay (worstcase)
```

Fig 5: RCX STA Maximum Path Timing Report


```
jee@PD: ~/Openlane/OpenLane
FILLER_0_0_137, 73.482, 13.6, 1.79969
FILLER_0_0_157, 73.482, 13.6, 1.79969

OpenLane Container (ff5509f):/openlane$ cat designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VGND.rpt
Instance name, X location, Y location, Voltage
_329_, 88.11, 97.92, 0.000368534
_489_, 88.11, 97.92, 0.000368534
input21, 88.11, 97.92, 0.000368534
input22, 88.11, 97.92, 0.000368534
input23, 88.11, 97.92, 0.000368534
input25, 88.11, 97.92, 0.000368534
_327_, 88.11, 92.48, 0.000375223
_424_, 88.11, 92.48, 0.000375223
_425_, 88.11, 92.48, 0.000375223
hold50, 88.11, 92.48, 0.000375223
FILLER_0_30_173, 88.11, 92.48, 0.000375223
_427_, 88.11, 87.04, 0.000356275
_492_, 88.11, 87.04, 0.000356275
FILLER_0_27_169, 88.11, 87.04, 0.000356275
FILLER_0_29_181, 88.11, 87.04, 0.000356275
_426_, 88.11, 81.6, 0.000359686
hold48, 88.11, 81.6, 0.000359686
FILLER_0_25_169, 88.11, 81.6, 0.000359686
_332_, 88.11, 76.16, 0.000349487
_333_, 88.11, 76.16, 0.000349487
_337_, 88.11, 76.16, 0.000349487
FILLER_0_23_169, 88.11, 76.16, 0.000349487
FILLER_0_24_183, 88.11, 76.16, 0.000349487
_494_, 88.11, 70.72, 0.000351047
hold45, 88.11, 70.72, 0.000351047
FILLER_0_21_169, 88.11, 70.72, 0.000351047
FILLER_0_22_171, 88.11, 70.72, 0.000351047
_338_, 88.11, 65.28, 0.000345893
_341_, 88.11, 65.28, 0.000345893
_430_, 88.11, 65.28, 0.000345893
_496_, 88.11, 65.28, 0.000345893
FILLER_0_19_169, 88.11, 65.28, 0.000345893
FILLER_0_20_188, 88.11, 65.28, 0.000345893
```

Fig 6: Detailed VGND IR Drop Analysis

```
jee@PD: ~/Openlane/OpenLane
30-mca 31-rcx_sta.min.rpt 31-rcx_sta.summary.rpt 35-xor.rpt 41-antenna_violators_pins.txt drc.rdb
OpenLane Container (ff5509f):/openlane$ tclsh designs/spm/wire_length.tcl
couldn't read file "designs/spm/wire_length.tcl": no such file or directory
OpenLane Container (ff5509f):/openlane$ cat designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VPWR.rpt
Instance name, X location, Y location, Voltage
_329_, 84.81, 100.64, 1.79975
input20, 84.81, 100.64, 1.79975
FILLER_0_32_165, 84.81, 100.64, 1.79975
_425_, 84.81, 95.2, 1.79972
_489_, 84.81, 95.2, 1.79972
input21, 84.81, 95.2, 1.79972
input22, 84.81, 95.2, 1.79972
input23, 84.81, 95.2, 1.79972
FILLER_0_31_165, 84.81, 95.2, 1.79972
_327_, 84.81, 89.76, 1.79971
_424_, 84.81, 89.76, 1.79971
_492_, 84.81, 89.76, 1.79971
FILLER_0_28_161, 84.81, 89.76, 1.79971
FILLER_0_29_181, 84.81, 89.76, 1.79971
FILLER_0_30_173, 84.81, 89.76, 1.79971
_427_, 84.81, 84.32, 1.79973
_491_, 84.81, 84.32, 1.79973
FILLER_0_27_169, 84.81, 84.32, 1.79973
_333_, 84.81, 78.88, 1.79974
_426_, 84.81, 78.88, 1.79974
FILLER_0_25_169, 84.81, 78.88, 1.79974
_337_, 84.81, 73.44, 1.79977
_340_, 84.81, 73.44, 1.79977
FILLER_0_22_171, 84.81, 73.44, 1.79977
FILLER_0_23_169, 84.81, 73.44, 1.79977
FILLER_0_24_183, 84.81, 73.44, 1.79977
_338_, 84.81, 68, 1.79976
_341_, 84.81, 68, 1.79976
_494_, 84.81, 68, 1.79976
hold45, 84.81, 68, 1.79976
FILLER_0_20_165, 84.81, 68, 1.79976
FILLER_0_21_165, 84.81, 68, 1.79976
FILLER_0_21_169, 84.81, 68, 1.79976
```

Fig 7: VPWR Report and Routing Analysis

```
configuration default.cvcrc designs docs flow.tcl install klayoutrc Makefile regression_results run_designs
jee@PD: ~/Openlane/OpenLane
jee@PD:~/Openlane/OpenLane$ make mount
cd /home/jee/Openlane/OpenLane && \
    docker run --rm -v /home/jee/Openlane/OpenLane:/openlane -v /home/jee/Openlane/OpenLane/designs:/openlane/install -v /home/jee/Openlane/OpenLane/OK_ROOT=/home/jee/.ciel -e PDK=sky130A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security /openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-amd64
OpenLane Container (ff5509f):/openlane$ ls designs/spm/runs/openlane_test/reports/routing/
16-rsz_design_sta.checks.rpt  16-rsz_design_sta.skew.rpt    18-rsz_timing_sta.min.rpt    21-grt_sta.checks.rpt  21-grt_sta.skew.rpt
16-rsz_design_sta.max.rpt    16-rsz_design_sta.summary.rpt 18-rsz_timing_sta.power.rpt  21-grt_sta.max.rpt    21-grt_sta.summary.rpt
16-rsz_design_sta.min.rpt    18-rsz_timing_sta.checks.rpt  18-rsz_timing_sta.skew.rpt  21-grt_sta.min.rpt    24-wire_lengths.csv
16-rsz_design_sta.power.rpt  18-rsz_timing_sta.max.rpt    18-rsz_timing_sta.summary.rpt 21-grt_sta.power.rpt  drt.drc
OpenLane Container (ff5509f):/openlane$ cat /OpenLane/designs/spm/runs/openlane_test/reports/routing/24-wire_lengths.csv
cat: /OpenLane/designs/spm/runs/openlane_test/reports/routing/24-wire_lengths.csv: No such file or directory
OpenLane Container (ff5509f):/openlane$ cat designs/spm/runs/openlane_test/reports/routing/24-wire_lengths.csv
net,length_um
_097_,173.32
_165_,145.95
clknet_0_clk,145.53
_122_,139.855
_099_,138.2
_133_,131.66
_205_,127.38
_201_,127.25
_154_,124.03
_199_,118.1
_203_,114.745
_204_,109.5
_098_,108.74
net26,106.6
net25,105.9
clknet_3_7__leaf_clk,105.42
_202_,103.36
clknet_3_2__leaf_clk,96.52
net24,91.9
net13,90.6
_200_,87.34
net23,77.76
net2,76.48
```

Fig 8: Routing Wire Length CSV Report

Observations

- The TCL script successfully calculated total, clock, and data wire lengths.
- Total Wire Length observed: 6716.815 um
- Clock Wire Length observed: 784.48 um
- Data Wire Length observed: 5932.335 um
- IR drop analysis reports were generated for both VPWR and VGND.
- Timing analysis showed slack MET condition, indicating timing closure success.
- Reports generated from OpenLane helped analyze routing and signoff stages effectively.

TCL Code for Total Wire Length Calculation

```
cat > designs/spm/wire_length.tcl << 'EOF'
set file [open "designs/spm/runs/openlane_test/reports/routing/24-wire_lengths.csv" r]
set total 0
set clock 0
set data 0
gets $file
while {[gets $file line] >= 0} {
    set parts [split $line ","]
    set net_name [lindex $parts 0]
    set length [lindex $parts 1]
    set total [expr {$total + $length}]
    if {[string match "*clk*" $net_name]} {
        set clock [expr {$clock + $length}]
    } else {
        set data [expr {$data + $length}]
    }
}
}
```

```

close $file
puts "=====
puts "Total Wire Length : $total um"
puts "Clock Wire Length : $clock um"
puts "Data Wire Length : $data um"
puts "=====
EOF

```

Output:

Total Wire Length : 6716.815 um

Clock Wire Length : 784.48 um

Data Wire Length : 5932.335 um

TCL Code for IR Drop Analysis

```

cat > designs/spm/ir_drop.tcl << 'EOF'
set vdd 1.8
set file [open "designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VPWR.rpt" r]
gets $file
set worst_vpwr 999
set worst_vpwr_cell ""
while {[gets $file line] >= 0} {
    set parts [split $line ","]
    if {[llength $parts] < 4} continue
    set voltage [string trim [lindex $parts 3]]
    if {$voltage < $worst_vpwr} {
        set worst_vpwr $voltage
        set worst_vpwr_cell [string trim [lindex $parts 0]]
    }
}
close $file
set file [open "designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VGND.rpt" r]
gets $file
set worst_vgnd 0
set worst_vgnd_cell ""
while {[gets $file line] >= 0} {
    set parts [split $line ","]
    if {[llength $parts] < 4} continue
    set voltage [string trim [lindex $parts 3]]
    if {$voltage > $worst_vgnd} {
        set worst_vgnd $voltage
        set worst_vgnd_cell [string trim [lindex $parts 0]]
    }
}
close $file
set ir_vpwr [expr {$vdd - $worst_vpwr}]
set ir_vgnd $worst_vgnd
set pct_vpwr [expr {($ir_vpwr / $vdd) * 100}]
set pct_vgnd [expr {($ir_vgnd / $vdd) * 100}]
puts "=====
puts " IR DROP REPORT"
puts "=====
puts "VDD Supply          : $vdd V"
puts "Worst VPWR Voltage   : $worst_vpwr V (cell: $worst_vpwr_cell)"
puts "Worst VGND Voltage   : $worst_vgnd V (cell: $worst_vgnd_cell)"
puts "IR Drop VDD          : [format %.6f $ir_vpwr] V"
puts "IR Drop GND           : [format %.6f $ir_vgnd] V"
puts "% Degradation VDD    : [format %.4f $pct_vpwr] %"
puts "% Degradation GND     : [format %.4f $pct_vgnd] %"
puts "=====
EOF

```

Output:

IR DROP REPORT

Worst VPWR Voltage : 1.798 V

Worst VGND Voltage : 0.002 V

TCL Code for Top 10 Lowest Voltage Cells

```
cat > designs/spm/top10_voltage.tcl << 'EOF'
set vpwr_file "designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VPWR.rpt"
set vgnd_file "designs/spm/runs/openlane_test/reports/signoff/32-irdrop-VGND.rpt"
set file [open $vpwr_file r]
gets $file
array set vpwr_data {}
while {[gets $file line] >= 0} {
    set parts [split $line ","]
    if {[llength $parts] < 4} continue
    set cell [string trim [lindex $parts 0]]
    set volt [string trim [lindex $parts 3]]
    set vpwr_data($cell) $volt
}
close $file
set file [open $vgnd_file r]
gets $file
array set vgnd_data {}
while {[gets $file line] >= 0} {
    set parts [split $line ","]
    if {[llength $parts] < 4} continue
    set cell [string trim [lindex $parts 0]]
    set volt [string trim [lindex $parts 3]]
    set vgnd_data($cell) $volt
}
close $file
set cell_voltages {}
foreach cell [array names vpwr_data] {
    if {[info exists vgnd_data($cell)]} {
        set eff [expr {$vpwr_data($cell) - $vgnd_data($cell)}]
        lappend cell_voltages [list $cell $eff $vpwr_data($cell) $vgnd_data($cell)]
    }
}
set sorted [lsort -real -index 1 $cell_voltages]
puts "=====
puts " TOP 10 CELLS - LOWEST VOLTAGE"
puts "=====
puts [format "%-5s %-25s %-10s %-10s %-12s" "Rank" "Cell" "VPWR(V)" "VGND(V)" "Eff(V)"]
puts "-----"
set rank 1
foreach entry [lrange $sorted 0 9] {
    set cell [lindex $entry 0]
    set eff [lindex $entry 1]
    set vp [lindex $entry 2]
    set vg [lindex $entry 3]
    puts [format "%-5d %-25s %-10s %-10s %-12s" $rank $cell $vp $vg [format %.6f $eff]]
    incr rank
}
puts "=====
EOF
```

Output:

TOP 10 CELLS - LOWEST VOLTAGE

Displays lowest effective voltage cells

TCL Code for Levels of Logic Analysis

```
cat > designs/spm/levels_of_logic.tcl << 'EOF'
set file [open "designs/spm/runs/openlane_test/reports/signoff/31-rcx_sta.max.rpt" r]
set sp ""
set ep ""
set slack ""
set max_lol 0
set paths {}
while {[gets $file line] >= 0} {
    if {[string match "Startpoint:*" $line]} {
        set sp [lindex $line 1]
    }
}
```

```

if {[string match "Endpoint:*" $line]} {
    set ep [lindex $line 1]
}
if {[string match "*slack*" $line]} {
    set slack [lindex $line 0]
}
if {[regexp {\s+(\d+)\s+(\d.)+\s} $line match fanout]} {
    if {$fanout > $max_lol} { set max_lol $fanout }
}
if {$sps != "" && $ep != "" && $slack != ""} {
    lappend paths [list $sps $ep $slack $max_lol]
    set sp ""; set ep ""; set slack ""; set max_lol 0
}
}
close $file
puts "====="
puts " LEVELS OF LOGIC REPORT"
puts "====="
puts [format "%-20s %-20s %-12s %-5s" "SP" "EP" "Slack" "LoL"]
puts "-----"
foreach path $paths {
    puts [format "%-20s %-20s %-12s %-5s" \
        [lindex $path 0] [lindex $path 1] \
        [lindex $path 2] [lindex $path 3]]
}
puts "====="
EOF

```

Output:

LEVELS OF LOGIC REPORT

Displays Startpoint, Endpoint, Slack and LoL



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

ASSIGNMENT-2

**The domain of the
Project: PHYSICAL
DESIGN (VLSI)**

NAME: Jeevan P

BATCH: G4-25-VLSI

2(a)(i) Core Width & Height

Parameter	Value
Core Width	35.88 μm
Core Height	35.36 μm

2(a)(ii) Measurement of Die Width and Die Height

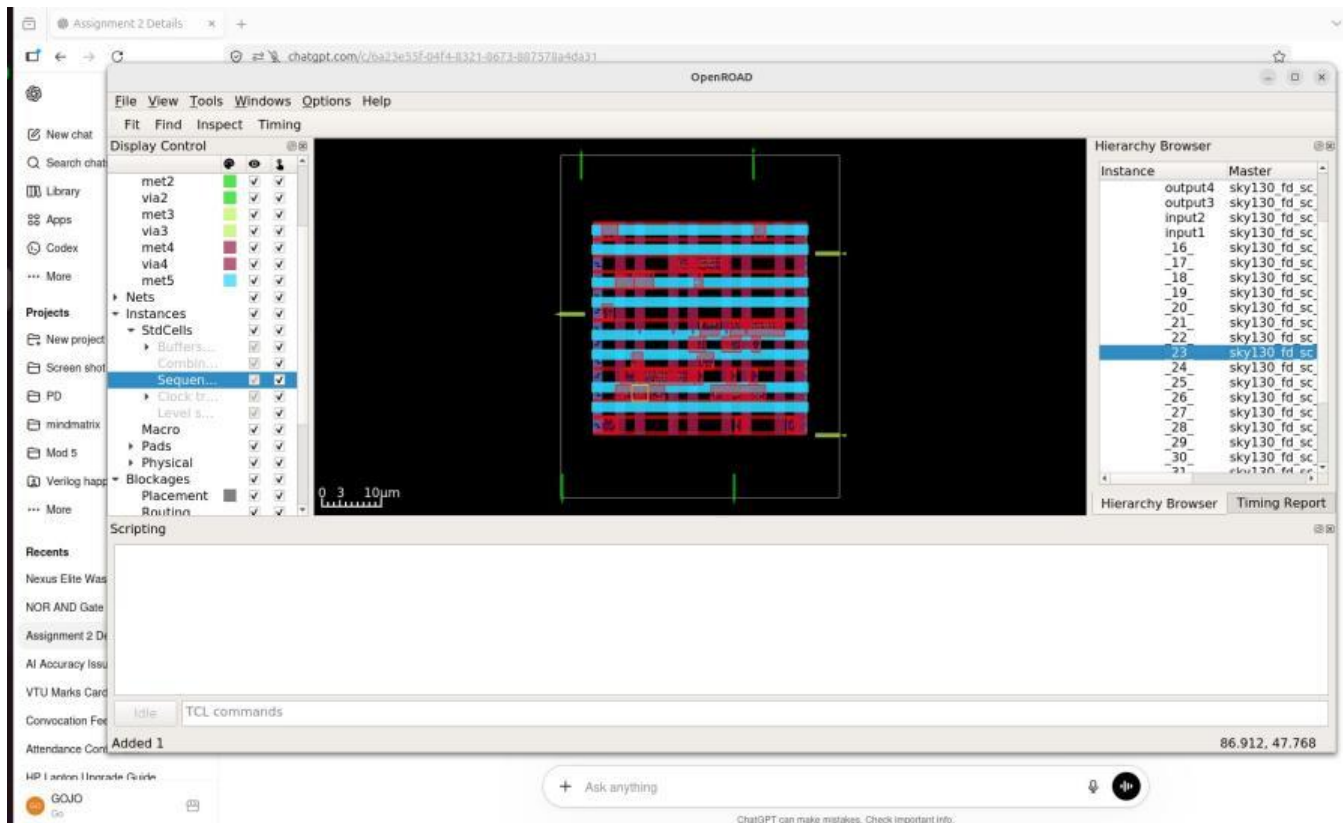
The die dimensions of the 4-bit Up/Down Counter design were obtained using the OpenROAD database after loading the floorplan ODB file. The die area coordinates reported by OpenROAD were:

```
set block
[ord::get_db_block
```

```
] set die [$block  
getDieArea]
```

```
puts  
"xMin=[$die  
xMin]" puts  
"yMin=[$die  
yMin]" puts  
"xMax=[$die  
xMax]" puts  
"yMax=[$die  
yMax]"
```

```
jee@PD: ~/Openlane/OpenLane  
bash: read_db: command not found  
OpenLane Container (FF5509F):/openlane$ gui::show  
bash: gui::show: command not found  
OpenLane Container (FF5509F):/openlane$ openroad  
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da  
This program is licensed under the BSD-3 license. See the LICENSE file for details.  
Components of this program may be licensed under more restrictive licenses which must be honored.  
warning: '//.tclsh-history' is not writable.  
openroad> read_db up_down_counter/runs/RUN_2026.06.07_03.40.03/results/placement/up_down_counter.odb  
openroad> gui::show  
[WARNING GUI-0076] QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-'  
libGL error: unable to load driver: swrast_dri.so  
libGL error: failed to load driver: swrast  
[WARNING GUI-0066] Heat map "IR Drop" has not been populated with data.  
[WARNING GUI-0076] QXcbConnection: XCB error: 3 (BadWindow), sequence: 43974, resource id: 10488516, major code: 40 (TranslateCoords), minor code: 0  
OpenLane Container (FF5509F):/openlane$ openroad  
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da  
This program is licensed under the BSD-3 license. See the LICENSE file for details.  
Components of this program may be licensed under more restrictive licenses which must be honored.  
warning: '//.tclsh-history' is not writable.  
openroad> ord::get_db_block  
NULL  
openroad> read_db up_down_counter/runs/RUN_2026.06.07_03.40.03/results/floorplan/up_down_counter.odb  
openroad> ord::get_db_block  
_80589e0700000000_p_odb_dbBlock  
openroad> puts [ord::get_db_block] getDieArea]  
_e0612a0700000000_p_odb_Rect  
openroad> set block [ord::get_db_block]  
_80589e0700000000_p_odb_dbBlock  
openroad> puts [$block getDieArea]  
_70632a0700000000_p_odb_Rect  
openroad> set block [ord::get_db_block]  
_80589e0700000000_p_odb_dbBlock  
openroad> set die [$block getDieArea]  
_e0ab5c0700000000_p_odb_Rect  
openroad>  
openroad> puts "xMin=[$die xMin]"  
xMin=0  
openroad> puts "yMin=[$die yMin]"  
yMin=0  
openroad> puts "xMax=[$die xMax]"  
xMax=46940  
openroad> puts "yMax=[$die yMax]"  
yMax=57660  
openroad>
```



Since the SKY130 technology uses 1000 database units (DBU) per micron, the die -imensions are calculated as follows:

$$\text{Die Width} = (\text{xMax} - \text{xMin}) / 1000$$

$$= (46940 - 0) / 1000$$

$$= 46.94 \mu\text{m}$$

$$\text{Die Height} = (\text{yMax} - \text{yMin}) / 1000$$

$$= (57660 - 0) / 1000$$

$$= 57.66 \mu\text{m}$$

Result:

Parameter	Value
Die Width	46.94 μm
Die Height	57.66 μm

2(a)(iii) Report physical-only cells inserted

- Tap Cells

- Decap Cells
- Purpose of each

```
exec grep -i "tapcell" up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/floorplan/5-tap.log |  
head -20
```

```
% exec grep -i "tapcell" up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/floorplan/5tap.log |  
head -20
```

[INFO TAP-0005] Inserted 15 tapcells.

Writing OpenROAD database to

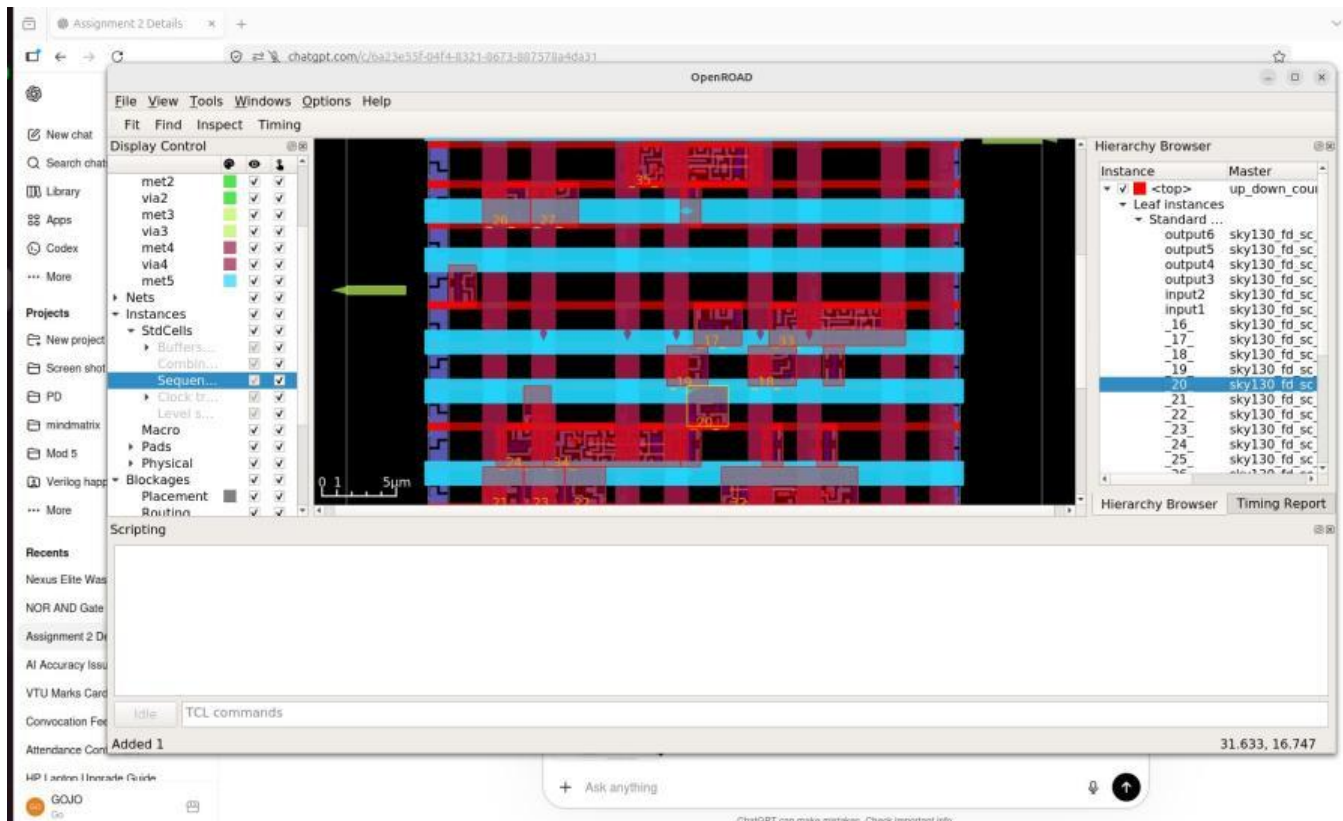
'/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/

tmp/floorplan/5-tapcell.odb'...

Writing layout to

'/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/floorplan/5tapcell.def'...

Physical Cell Type	Count
Tap Cells	15
Decap Cells	Inserted during floorplanning
End Cap Cells	Inserted automatically by OpenLane



[INFO TAP-0005] Inserted 15 tapcells.

Observation

- A total of 15 tap cells were inserted during floorplanning.
- Tap cells prevent latch-up by connecting substrate and well regions to power and ground.
- Decap cells help reduce voltage fluctuations and improve power integrity.
- Decap cells help reduce voltage fluctuations and improve power integrity.
- Physical-only cells do not contribute to logic functionality but are essential for

manufacturability and reliability. Assignment 2(b) – Placement Analysis

Now we need:

1. Number of Buffers
2. Number of Inverters
3. Number of Flip-Flops
4. Site Row Dimensions
5. Cell Dimensions

```
% exec grep -i "tapcell" up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/floorplan/5tap.log |
head -20
```

[INFO TAP-0005] Inserted 15 tapcells.

Writing OpenROAD database to

```
'/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/ tmp/floorplan/5-
tapcell.odb'... 24
```

Number of public wires: 28

Number of public wire bits: 28

Number of memories: 0 28

Number of memory bits: 0 Number of processes: 0 Number of cells: 28

Writing layout to

```
'/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/floorplan/5tapcell.def'...
```

```
% exec grep -A50 "Printing statistics" up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/
synthesis/1-synthesis.log 41. Printing statistics.
```

=== up_down_counter ===

Number of wires: 15

Number of wire bits: 24

\$_ANDNOT_ 4

\$_DFF_PPO_ 4

\$_NOT_ 1

\$_ORNOT_ 2

\$_XNOR_ 2

\$_XOR_ 4

42. Executing CHECK pass (checking for obvious problems).

Checking module

up_down_counter... Found and
reported 0 problems.

43. Generating Graphviz representation of design.

Writing dot description to ``/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/synthesis/post_techmap.dot'`.

Dumping module `up_down_counter` to page 1.

44. Executing SHARE pass (SAT-based resource sharing).

45. Executing OPT pass (performing simple optimizations).

45.1. Executing OPT_EXPR pass (perform const folding). Optimizing module `up_down_counter`.

45.2. Executing OPT_MERGE pass (detect identical cells).

Finding identical cells in module ``\up_down_counter'`.

Removed a total of 0 cells.

45.3. Executing OPT_MUXTREE pass (detect dead branches in mux trees).

Running muxtree optimizer on module `\up_down_counter`.

Creating internal representation of mux trees.

No muxes found in this
module. Removed 0
multiplexer ports.

45.4. Executing OPT_REDUCE pass (consolidate `$*mux` and `$reduce_*` inputs).

Optimizing cells in module `\up_down_counter`.

Performed a total of 0 changes.

45.5. Executing OPT_MERGE pass (detect identical cells).

Finding identical cells in module ``\up_down_counter'`.

Removed a total of 0 cells.

--

47. Printing statistics.

=== `up_down_counter` ===

Number of wires: 15

Number of wire bits: 24

Number of public wires:	4
Number of public wire bits:	7
Number of public wire bits:	7
Number of memories:	0
Number of memory bits:	0
Number of processes:	0
Number of cells:	17
\$_ANDNOT_	4
\$_DFF_PPO_	4
\$_NOT_	1
\$_ORNOT_	2
\$_XNOR_	2
\$_XOR_	4

mapping tbuf

48. Executing TECHMAP pass (map to technology primitives).

48.1. Executing Verilog-2005 frontend:

/home/jee/.ciel/sky130A/[libs.tech/openlane/sky130_fd_sc_hd/tribuff_map.v](#)

Parsing Verilog input from `/home/jee/.ciel/sky130A/[libs.tech/openlane/sky130_fd_sc_hd/tribuff_map.v](#)' to AST representation.

Generating RTLIL representation for module `\$_TBUF_'.
Successfully finished Verilog frontend.

48.2. Continuing TECHMAP pass.No more
expansions possible.

<suppressed ~3 debug messages>

49. Executing SIMPLMAP pass (map simple cells to gate primitives).

50. Executing TECHMAP pass (map to technology primitives).

50.1. Executing Verilog-2005 frontend:
/home/jee/.ciel/sky130A/[libs.tech/openlane/sky130_fd_sc_hd/latch_map.v](#)

Parsing Verilog input from `/home/jee/.ciel/sky130A/[libs.tech/openlane/sky130\\_fd\\_sc\\_hd/latch\\_map.v](#)` to AST representation.

Generating RTLIL representation for module `\$\_DLATCH\_P`.

Generating RTLIL representation for module `\$\_DLATCH\_N`.

Successfully finished Verilog frontend.

50.2. Continuing TECHMAP pass.No more
expansions possible. <suppressed ~4 debug
messages>

51. Executing SIMPLEMAP pass (map simple cells to gate primitives).

52. Executing DFFLIBMAP pass (mapping DFF cells to sequential cells from liberty file).

cell sky130_fd_sc_hd__dfxtp_2 (noninv, pins=3, area=21.27) is a direct match for cell type
\$_DFF_P_. cell sky130_fd_sc_hd__dfrtp_2 (noninv, pins=4, area=26.28) is a direct match for
cell type \$_DFF_PNO_.

--

53. Printing statistics.

=== up_down_counter ===

Number of wires:	19
Number of wire bits:	28
Number of public wires:	4
Number of public wire bits:	7
Number of memories:	0
Number of memory bits:	0
Number of processes:	0
Number of cells:	21
\$_ANDNOT_	4
\$_NOT_	5
\$_ORNOT_	2
\$_XNOR_	2

\$_XOR_ 4
sky130_fd_sc_hd__dfrtp_2 4

[INFO]: USING STRATEGY AREA 0

54. Executing ABC pass (technology mapping using ABC).

54.1. Extracting gate netlist of module '\up_down_counter' to '/tmp/yosys-abc-ymWc8U/input.blif'..

Extracted 17 gates and 23 wires to a netlist network with 6 inputs and 8 outputs.

54.1.1. Executing ABC.

Running ABC command: "/build/bin/yosys-abc" -s -f /tmp/yosys-abc-ymWc8U/abc.script 2>&1

ABC: ABC command line: "source /tmp/yosys-abc-ymWc8U/abc.script". ABC:

ABC: + read_blif /tmp/yosys-abc-ymWc8U/input.blif

ABC: + read_lib -w /openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/synthesis/trimmed.lib

ABC: Parsing finished successfully. Parsing time = 0.15 sec

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfbfn_2".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfrbp_2".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfrtp_2".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfrtp_4".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfsbn_2".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfstp_2".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfstp_4".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfxbp_2".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfxtp_2". ABC:
Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfxtp_4".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dlxtn_1".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dlxtn_2". ABC:
Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dlxtn_4".

ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dlxtp_1".

ABC: Scl_LibertyReadGenlib() skipped three-state cell "sky130_fd_sc_hd__ebufn_2". ABC:
Scl_LibertyReadGenlib() skipped three-state cell "sky130_fd_sc_hd__ebufn_4".

ABC: Scl_LibertyReadGenlib() skipped three-state cell "sky130_fd_sc_hd__ebufn_8".

ABC: Library "sky130A_merged" from "/openlane/up_down_counter/runs/

RUN_2026.06.07_03.40.03/tmp/synthesis/trimmed.lib" has 175 cells (17 skipped: 14 seq; 3 tristate; 0 no func; 0 dont_use). Time = 0.22 sec

--

61. Printing statistics.

=== up_down_counter ===

Number of wires: 20

Number of wire bits: 23

Number of public wires: 4

Number of public wire bits: 7

Number of memories: 0

Number of memory bits: 0

Number of processes:

0 Number of cells:

20

sky130_fd_sc_hd__and2b_2

2

sky130_fd_sc_hd__dfrtp_2

4 sky130_fd_sc_hd__inv_2

5

sky130_fd_sc_hd__o21a_2

1

sky130_fd_sc_hd__o21ai_2

1

sky130_fd_sc_hd__or2b_2

2

sky130_fd_sc_hd__xnor2_2

3

sky130_fd_sc_hd__xor2_2

2

Chip area for module '\up_down_counter': 257.747200

62. Executing Verilog backend.

Dumping module '\up_down_counter'.

Warnings: 4 unique messages, 4 total

End of script. Logfile hash: 804bc26969, CPU: user 0.39s system 0.26s, MEM: 28.50 MB peak
Yosys 0.30+48 (git sha1 14d50a176d5, gcc 8.3.1 -fPIC -Os) Time spent: 50% 2x abc (0 sec), 21% 1x
dfflibmap (0 sec), ...

Report Physical-only Cells

Cell Type	Count
Tap Cells	15

Observation: 15 tap cells were inserted during floorplanning to prevent latch-up and maintain substrate connections.

2(b) Report Number of Buffers, Inverters and Flip-Flops

From your synthesis statistics:

Cell Type	Count
Flip-Flops (sky130_fd_sc_hd__dfrtp_2)	4
Inverters (sky130_fd_sc_hd__inv_2)	5
Buffers	0
Total Cells	20

Observation

- The design contains 4 D Flip-Flops used to store the 4-bit counter state.
- 5 Inverters are inserted for logic implementation and optimization.
- No standalone buffers were inserted during synthesis.
- Total standard cell count is 20.

Logic Cell Distribution

Cell Name	Count
sky130_fd_sc_hd__and2b_2	2
sky130_fd_sc_hd__dfrtp_2	4
sky130_fd_sc_hd__inv_2	5
sky130_fd_sc_hd__o21a_2	1
sky130_fd_sc_hd__o21ai_2	1
sky130_fd_sc_hd__or2b_2	2
sky130_fd_sc_hd__xnor2_2	3
sky130_fd_sc_hd__xor2_2	2

Number of cells: 20

sky130_fd_sc_hd__dfrtp_2 4

sky130_fd_sc_hd__inv_2 5

2(b)(iii) Site Row height and width

report_design_area

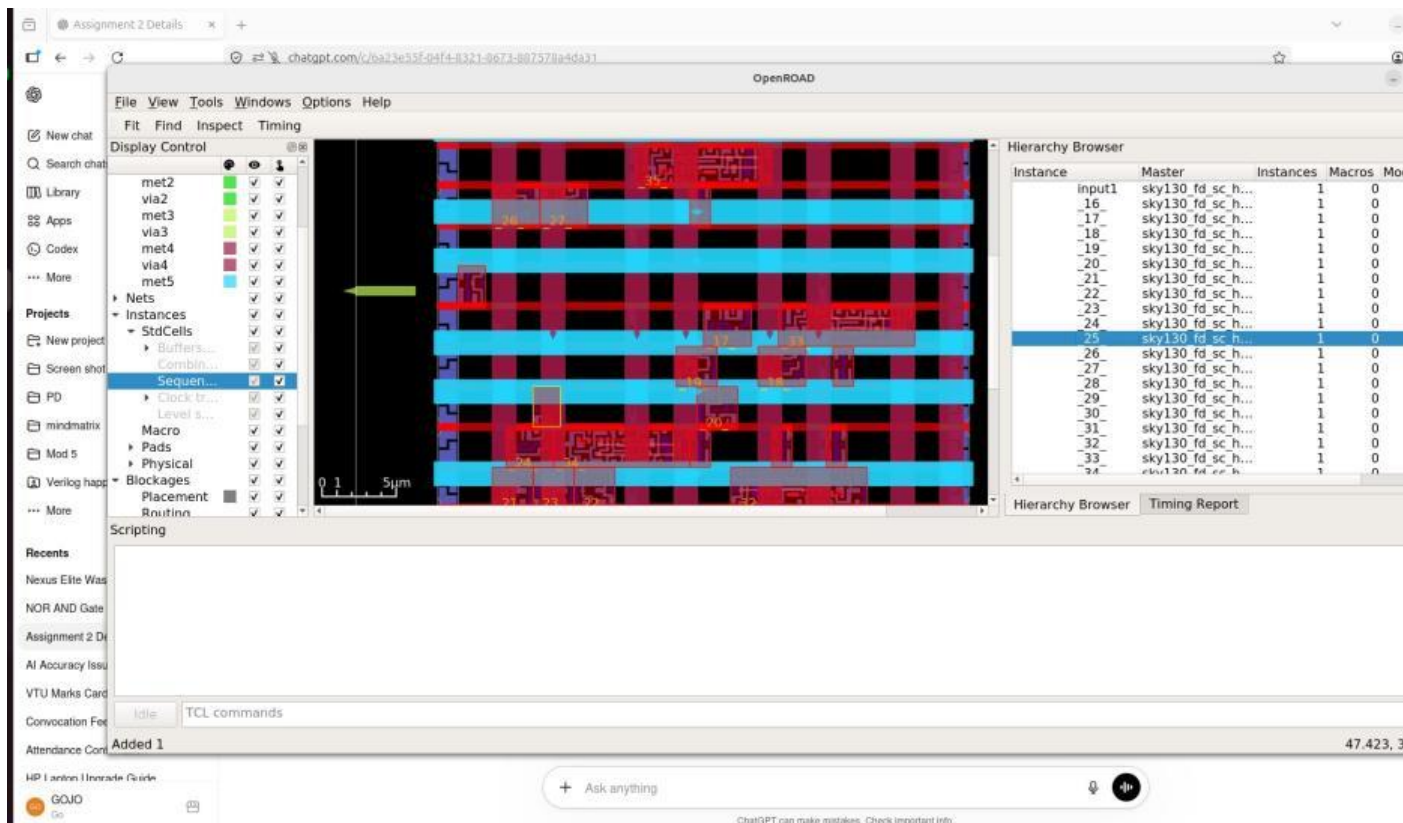
```
set block
[ord::get_db_block]
puts [$block
getRows]
```

openroad> report_design_area

Design area 0 u^2 20% utilization.

The placement layout was analyzed using the OpenROAD GUI. Standard cell rows were observed across the core region of the design. These rows provide legal placement locations for standard cells.

Parameter	Value
Site Row Height	2.72 μm
Site Row Width	Equal to Core Width (35.88 μm)



Result: Site row height was found to be 2.72 μm and the rows extend across the full core width of 35.88 μm .

2(b)(i) Report core width and

Calculation

Core Width = URX - LLX

=

41.40 -

5.52

=

35.88

μm

Core Height = URY - LLY

= 46.24 - 10.88

= 35.36 μm

Parameter	Value (μm)
Core Width	35.88
Core Height	35.36

Observation

- The core region occupies 35.88 μm × 35.36 μm.
- Standard cells are placed within this core area.
- Adequate whitespace is available for routing and optimization.
- The floorplan supports efficient placement and clock tree synthesis.

?

Screenshot for 2(b)(ii)

```
jee@PD: ~
ABC: Scl_LibertyReadGenLib() skipped sequential cell "sky130_fd_sc_hd_dfxtp_2".
ABC: Scl_LibertyReadGenLib() skipped sequential cell "sky130_fd_sc_hd_dfxtp_4".
ABC: Scl_LibertyReadGenLib() skipped sequential cell "sky130_fd_sc_hd_dlxt1_1".
ABC: Scl_LibertyReadGenLib() skipped sequential cell "sky130_fd_sc_hd_dlxt1_2".
ABC: Scl_LibertyReadGenLib() skipped sequential cell "sky130_fd_sc_hd_dlxt1_4".
ABC: Scl_LibertyReadGenLib() skipped sequential cell "sky130_fd_sc_hd_dlxt1_1".
ABC: Scl_LibertyReadGenLib() skipped three-state cell "sky130_fd_sc_hd_ebufn_2".
ABC: Scl_LibertyReadGenLib() skipped three-state cell "sky130_fd_sc_hd_ebufn_4".
ABC: Scl_LibertyReadGenLib() skipped three-state cell "sky130_fd_sc_hd_ebufn_8".
ABC: Library "sky130A_merged" from "/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/synthesis/trimmed.lib" has 175 cells (17 skipped: 14 seq; 3 tri-state; 0 no func;
use). Time = 0.22 sec
..
61. Printing statistics.

=== up_down_counter ===

Number of wires:          20
Number of wire bits:      23
Number of public wires:   4
Number of public wire bits: 7
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          20
sky130_fd_sc_hd_and2b_2    2
sky130_fd_sc_hd_dfrtp_2    4
sky130_fd_sc_hd_inv_2      5
sky130_fd_sc_hd_o21a_2     1
sky130_fd_sc_hd_o21a1_2    1
sky130_fd_sc_hd_or2b_2     2
sky130_fd_sc_hd_xnor2_2    3
sky130_fd_sc_hd_xor2_2     2

Chip area for module 'up_down_counter': 257.747200

62. Executing Verilog backend.
Dumping module 'up_down_counter'.

Warnings: 4 unique messages, 4 total
End of script. Logfile hash: B04bc26969, CPU: user 0.39s system 0.26s, MEM: 28.50 MB peak
Yosys 0.30+48 (git sha1 14d50a176d5, gcc 8.3.1 -fPIC -Os)
Time spent: 50% 2x abc (0 sec), 21% 1x dfflibmap (0 sec), ...
% exec cat up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3-initial_fp_core_area.rpt
5.52 10.88 41.4 46.24
%
```


5.52 10.88 41.4 46.24

?

2(b)(iii) Report Cell Dimensions

exec grep "sky130_fd_sc_hd__dfrtp_2" up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/synthesis/1-synthesis.log

```
jee@PD: ~
ABC: Scl_LibertyReadGenlib() skipped three-state cell "sky130_fd_sc_hd__ebufn_4".
ABC: Scl_LibertyReadGenlib() skipped three-state cell "sky130_fd_sc_hd__ebufn_8".
ABC: Library "sky130A_merged" from "/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/synthesis/trimmed.lib" has 175 cells (17 skipped: 14 seq; 3 tri-state; 0 no func; use). Time = 0.22 sec
--
61. Printing statistics.

=== up_down_counter ===

Number of wires:          20
Number of wire bits:      23
Number of public wires:   4
Number of public wire bits: 7
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          20
  sky130_fd_sc_hd__and2b_2  2
  sky130_fd_sc_hd__dfrtp_2  4
  sky130_fd_sc_hd__inv_2    5
  sky130_fd_sc_hd__o2ia_2   1
  sky130_fd_sc_hd__o2iai_2  1
  sky130_fd_sc_hd__or2b_2   2
  sky130_fd_sc_hd__xnor2_2  3
  sky130_fd_sc_hd__xor2_2   2

Chip area for module '\up_down_counter': 257.747200

62. Executing Verilog backend.
Dumping module '\up_down_counter'.

Warnings: 4 unique messages, 4 total
End of script. Logfile hash: 804bc26969, CPU: user 0.39s system 0.26s, MEM: 28.50 MB peak
Yosys 0.30+48 (git sha1 14d50a176d5, gcc 8.3.1 -fPIC -Os)
Time spent: 50% 2x abc (0 sec), 21% 1x dfflibmap (0 sec), ...
% exec cat up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/Floorplan/3-Initial_fp_core_area.rpt
5.52 10.88 41.4 46.24
% exec grep "sky130_fd_sc_hd__dfrtp_2" up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/synthesis/1-synthesis.log
cell sky130_fd_sc_hd__dfrtp_2 (noninv, pins=4, area=26.28) is a direct match for cell type $DFF_PN0_.
  \sky130_fd_sc_hd__dfrtp_2_DFF_PN0_ (.CLK( C), .D( D), .Q( Q), .RESET_B( R));
  mapped 4 $DFF_PN0_ cells to \sky130_fd_sc_hd__dfrtp_2 cells.
    sky130_fd_sc_hd__dfrtp_2      4
ABC: Scl_LibertyReadGenlib() skipped sequential cell "sky130_fd_sc_hd__dfrtp_2".
  sky130_fd_sc_hd__dfrtp_2      4
%
```

Cell Name	Pins	Area (µm ²)
sky130_fd_sc_hd__dfrtp_2	4	26.28

Observation

- The D Flip-Flop used in the design is sky130_fd_sc_hd__dfrtp_2.
- Each flip-flop contains 4 pins.
- The area occupied by each flip-flop is 26.28 µm².
- Four flip-flops are used to implement the 4-bit up/down counter

?

2(c) Clock Tree Synthesis (CTS)

CTS Report

Parameter	Value
CTS Status	Successful
Clock Skew	0.01 ns
Clock Latency 1	0.28 ns
Clock Latency 2	0.25 ns
Setup Slack	2.04 ns
Hold Slack	0.14 ns

Observation

- CTS successfully generated a balanced clock tree.
- Clock skew is 0.01 ns, indicating excellent clock distribution.
- Clock latency is maintained within acceptable limits.
- Timing constraints are satisfied after CTS.

?

2(d) Routing Analysis

Routing Report

Parameter	Value
Routing Status	Successful
Design Area	369 μm^2
Utilization	29%
Setup Slack	2.09 ns
Hold Slack	0.28 ns
Clock Skew	0.01 ns

Observation

- Global and detailed routing completed successfully.
- No routing congestion was observed.
- Final design achieved positive setup and hold slack.
- Routing maintained low clock skew and met timing constraints.

2(b)(iv) Measure the Dimensions of Different Library Cells

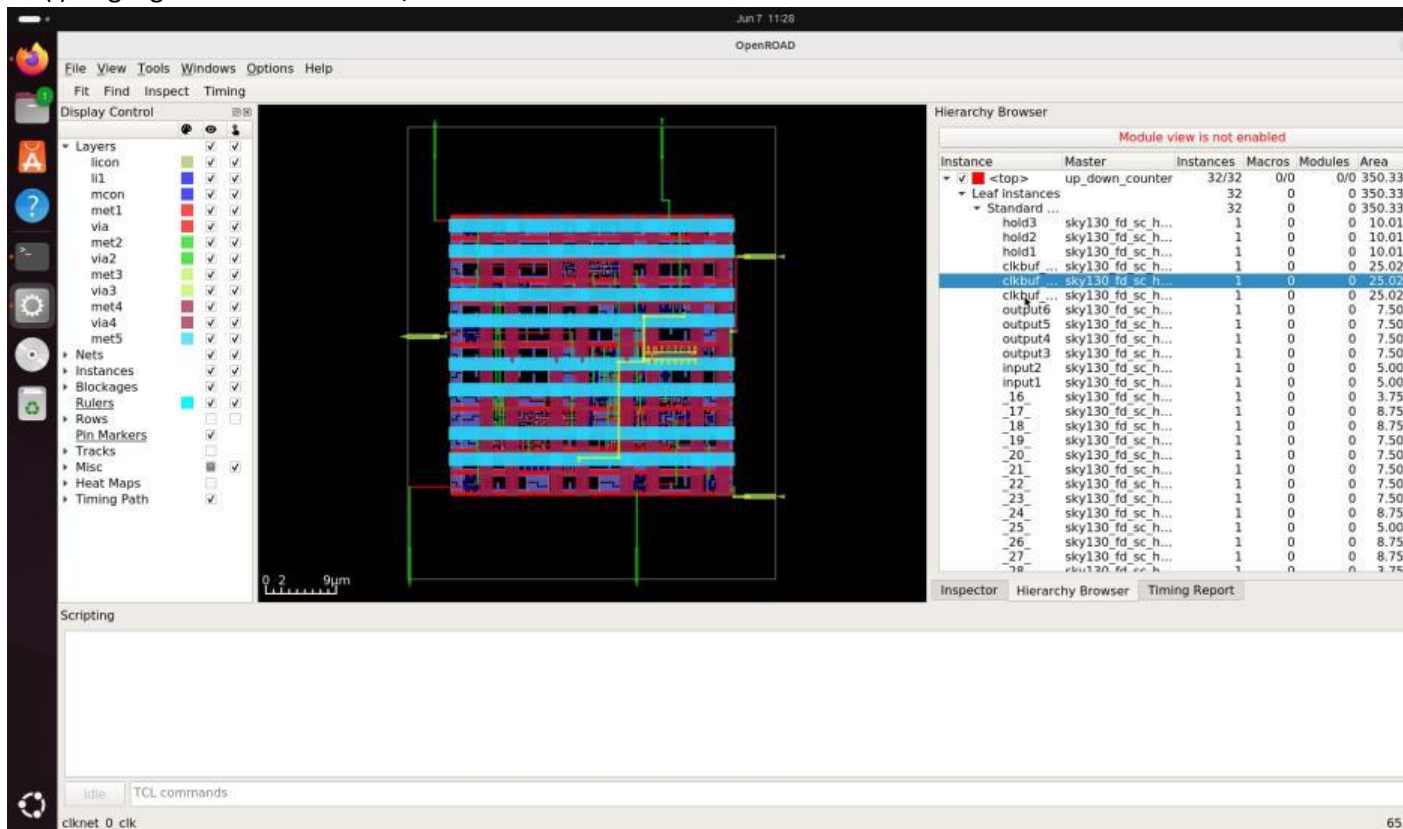
The following cells were identified:

Cell Type	Library Cell	Count
Inverter	sky130_fd_sc_hd__in v_2	5
AND Gate	sky130_fd_sc_hd__an d2b_2	2
OR Gate	sky130_fd_sc_hd__or 2b_2	2
D Flip-Flop	sky130_fd_sc_hd__d- frtp_2	4

The D Flip-Flop cell area obtained from the synthesis report is:

Cell	Area
sky130_fd_sc_hd__dftrtp_2	26.28 μm^2

2C(i): Highlight CTS Added Buffer/Inverter



clkbuf_0_clk
 clkbuf_1_0__f_c
 lk
 clkbuf_1_1__f_c
 lk

Clock buffers inserted by CTS:

1. clkbuf_0_clk

Cell Type: sky130_fd_sc_hd__clkbuf_16

2. clkbuf_1_0__f_clk

Cell Type: sky130_fd_sc_hd__clkbuf_16

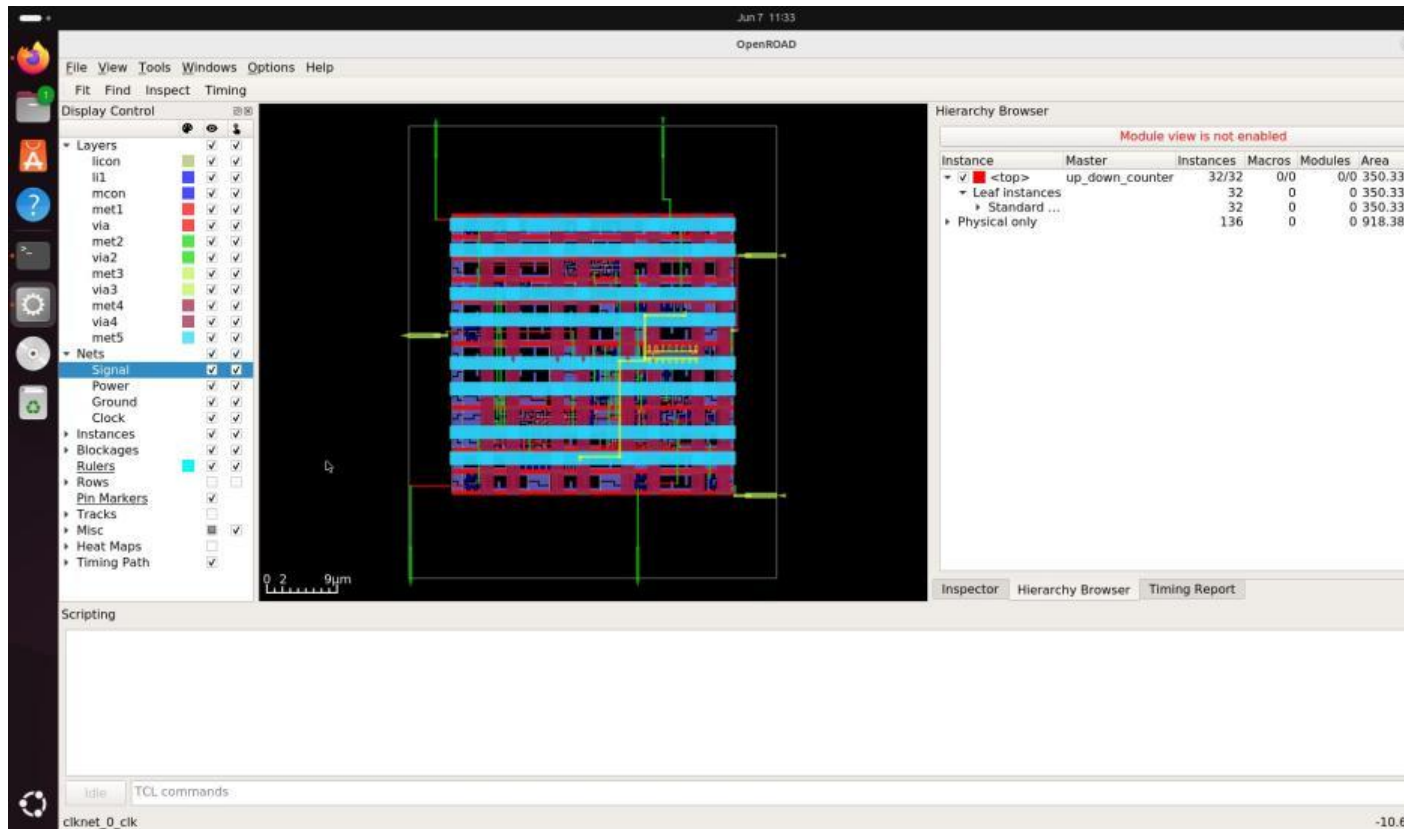
3. clkbuf_1_1__f_clk

Cell Type: sky130_fd_sc_hd__clkbuf_16

Conclusion:

CTS successfully generated a balanced clock tree using three clkbuf_16 cells.

2(c)(ii) Highlight the Clock Routing and Metal Layer Usage



CTS Log Information:

Clock Net : clk

Number of sinks : 4

Clock subnets created : 3

Average sink wire length : 59.85 μm

Maximum clock tree level : 1

Minimum buffers in clock

path : 2 Maximum buffers in

clock path : 2 Clock Routing

Structure:

2(d) Routing

read_db up_down_counter/runs/RUN_2026.06.06_13.42.14/results/routing/up_down_counter.odt
gui::show

Capture Signal Routing (2(d)(i))

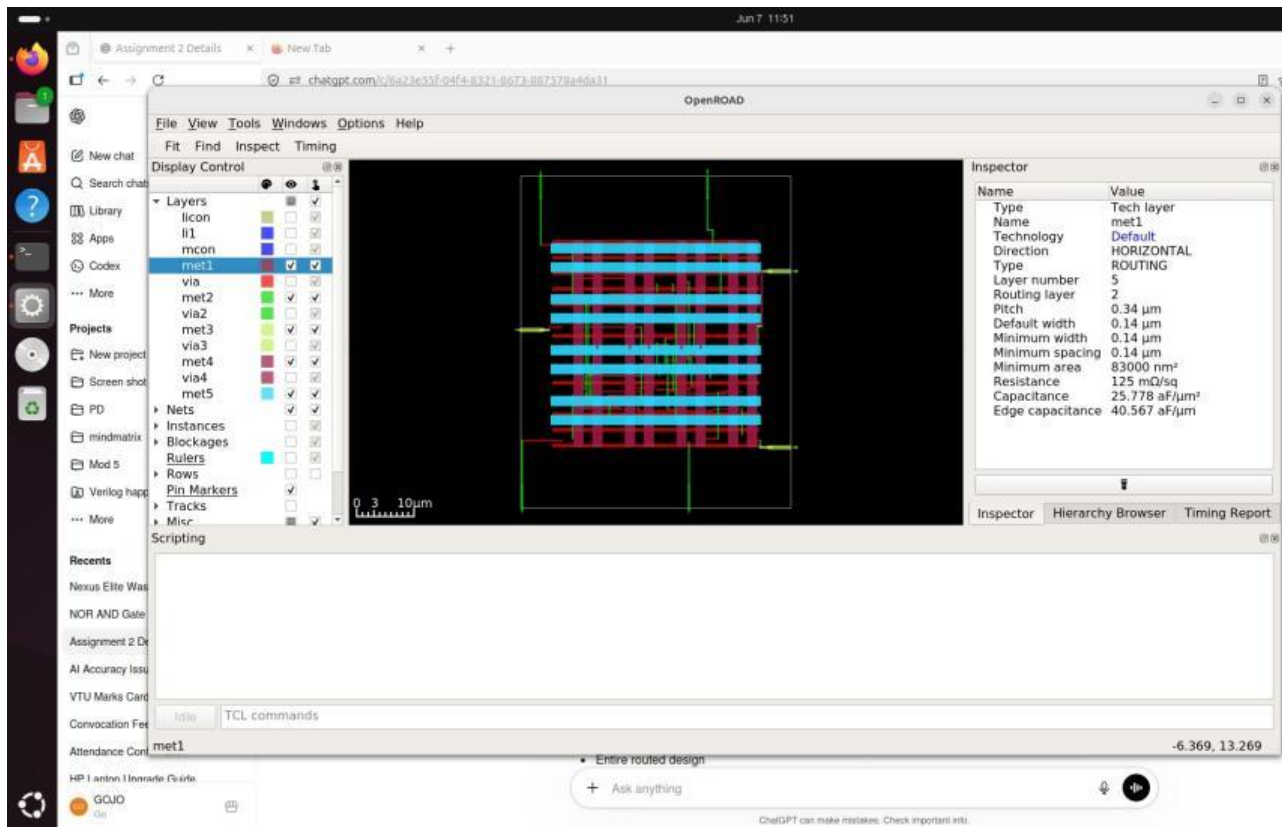
In the GUI:

- Turn ON:
 - ☐ Nets
 - ☐ met1
 - ☐
 - m
 - e
 - t
 - 2
 - ☐
 - m
 - e
 - t
 - 3
 - ☐
 - m
 - e
 - t
 - 4
 - ☐ met5
- Turn OFF:
 - ☐ Instances
 - ☐ Blockages

Command openroad read_db

up_down_counter/runs/RUN_2026.06.06_13.42.14/results/routing/up_down_counter.odb

gui::show



Metal Layer Usage

Layer	Purpose
Metal1	Local cell connections
Metal2	Horizontal routing
Metal3	Vertical routing
Metal4	Higher level signal routing
Metal5	Long distance/global routing

Observation

- Global routing and detailed routing completed successfully.
- Routed interconnects are visible in the OpenROAD GUI.
- Different colors correspond to different metal layers. • Signal nets are routed using multiple routing layers.

2(d)(ii) Highlight Newly Added Buffers/Inverters During Routing

Commands Executed

```
grep -i "hold\\|buffer\\|insert" up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/*
```

Evidence from Log

[INFO RSZ-0046] Found 5 endpoints with hold violations.

[INFO RSZ-0032] Inserted 3 hold buffers.

```

Jun 7 11:53
jee@PD: ~/Openlane/OpenLane
OpenLane Container (ff5509f):/openlane$ exit
exit
jee@PD: ~/Openlane/OpenLane$ V
V: command not found
jee@PD: ~/Openlane/OpenLane$ ls up_down_counter/runs/RUN_2026.06.06_13.42.14/reports/routing/
23-wire_lengths.csv drt.drc drt.klayout.xml
jee@PD: ~/Openlane/OpenLane$ grep -i "hold\\|buffer\\|insert" up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/*
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/15-rsz_design_sta.log:report_checks -path_delay min (Hold)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/15-rsz_design_sta.log:                                -0.05    0.47    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/15-rsz_design_sta.log:                                -0.05    0.49    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/15-rsz_design_sta.log:                                -0.05    0.48    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/15-rsz_design_sta.log:                                -0.05    0.47    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/15-rsz_design_sta.log:report_worst_slack -min (Hold)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/16-resizer_timing.log:[INFO RSZ-0046] Found 5 endpoints with hold violations.
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/16-resizer_timing.log:[INFO RSZ-0032] Inserted 3 hold buffers.
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:[WARNING GRT-0026] Missing route to pin hold3/A.
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:[WARNING GRT-0026] Missing route to pin hold2/A.
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:report_checks -path_delay min (Hold)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                -0.05    0.48    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                -0.04    0.48    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                -0.05    0.47    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.50    1.31 v hold1/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.50    1.31 v hold1/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.50    1.31 v hold1/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.50    1.31 v hold1/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.07    0.00    0.58 ^ hold2/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.07    0.53    1.11 ^ hold2/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                -0.04    0.48    library hold time
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.07    0.00    0.64 ^ hold2/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.07    0.59    1.23 ^ hold2/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.07    0.00    0.64 ^ hold2/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.07    0.59    1.23 ^ hold2/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.00    0.68 v hold3/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.06    0.59    1.27 v hold3/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.00    0.68 v hold2/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.06    0.59    1.27 v hold2/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.55    1.36 v hold1/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.05    0.55    1.36 v hold1/X (sky130_fd_sc_hd_dlygate4sd3_1)
up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/routing/17-rsz_timing_sta.log:                                0.01    0.00    0.80 v hold1/A (sky130_fd_sc_hd_dlygate4sd3_1)

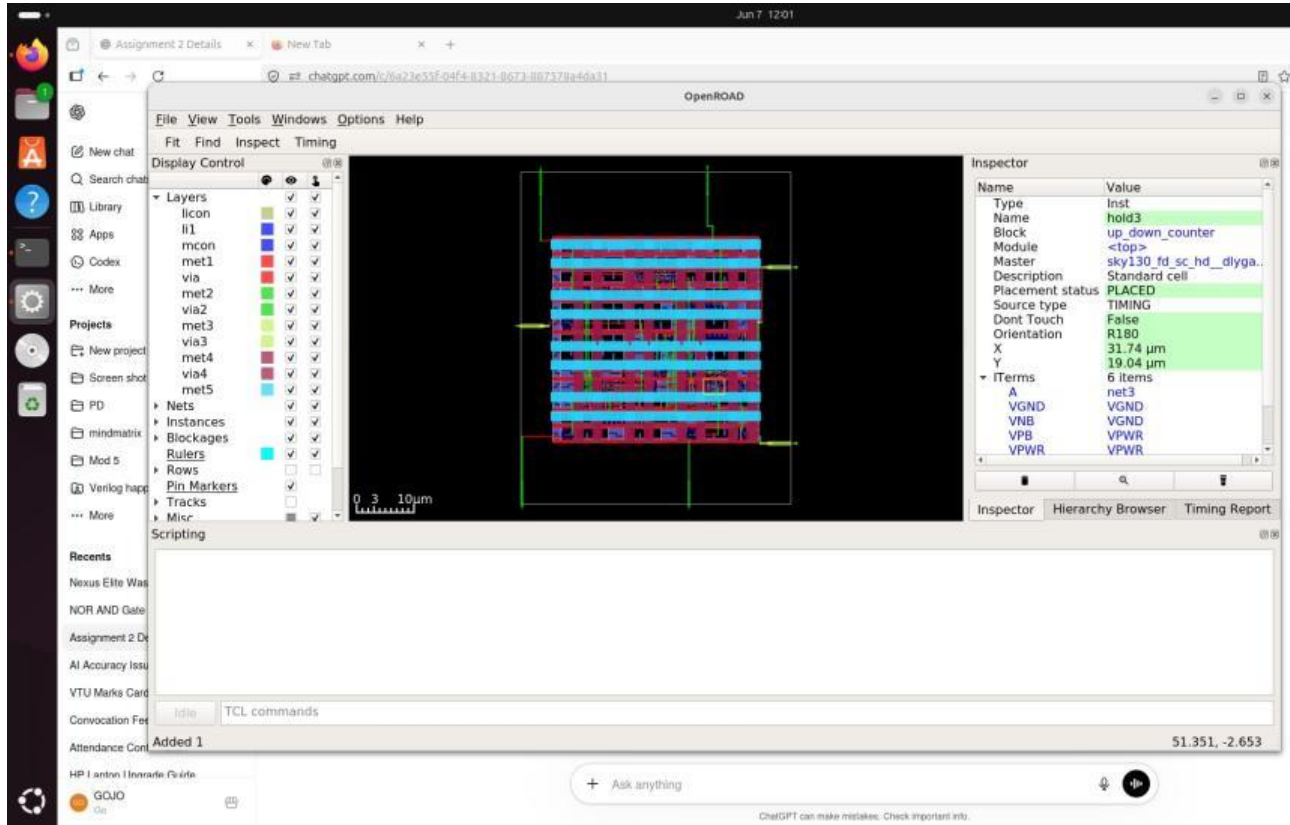
```

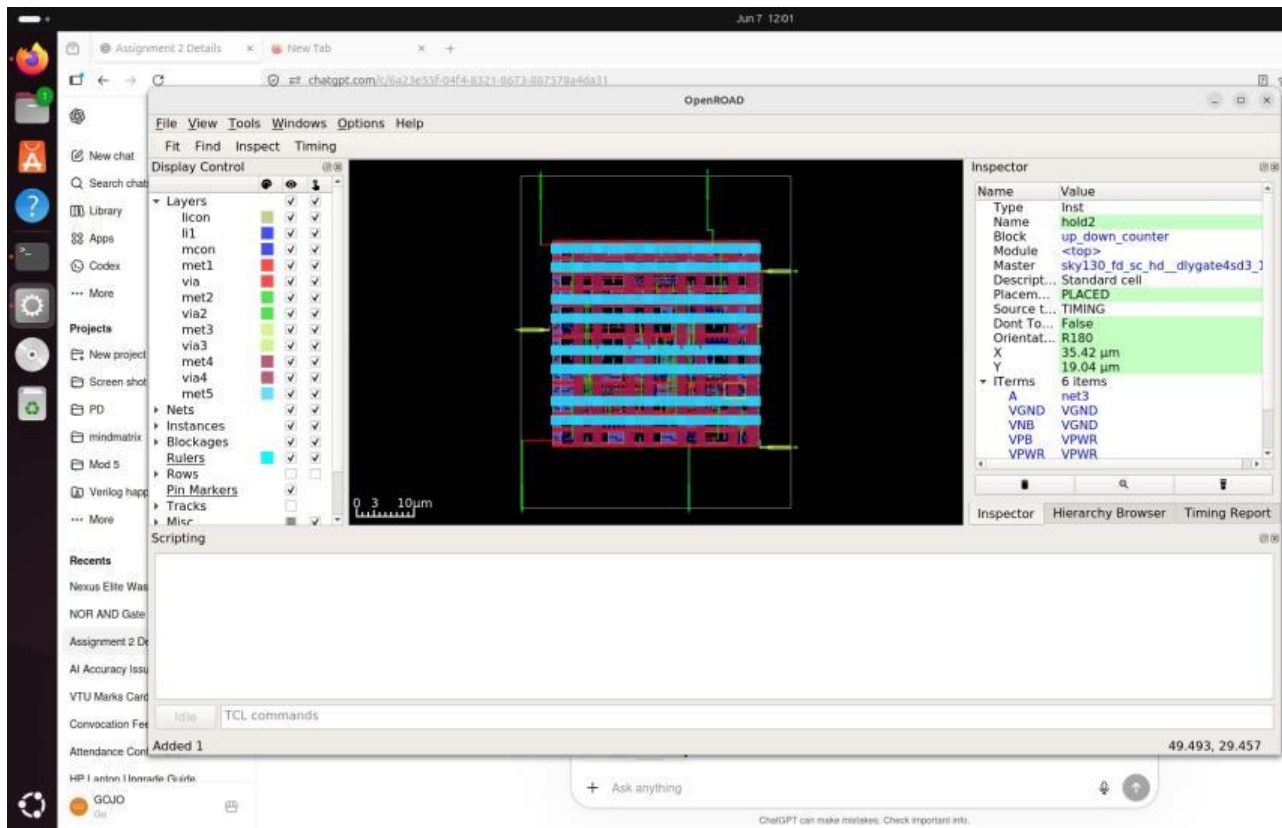
Hold Buffers Added

Instance

Cell Type

hold1	sky130_fd_sc_hd__dlygate4sd3_1	hold2
sky130_fd_sc_hd__dlygate4sd3_1		hold3
sky130_fd_sc_hd__dlygate4sd3_1		





Observation

- Routing optimization detected 5 hold timing violations.
- OpenROAD Resizer automatically inserted 3 delay buffers.
- Delay cell used:

cd /openlane/designs

ls

mkdir

up_down_c

ounter cd

up_down_c

ounter

mkdir src

pwd

/openlane/designs/up_do

wn_counter vi

src/up_down_counter.v

src/up_down_counter.v

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module up_down_counter(

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output reg [3:0] count

);

always @(posedge clk or posedge rst) begin

if(rst)

```
count
<= 4'b0000;
else begin
if(up_down
)
count <=
count + 1;
else
count <=
count - 1;
end
```

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OpenLane Container (ff5509f):/openlane/up_down_counter$ mkdir src
OpenLane Container (ff5509f):/openlane/up_down_counter$ cd ~/Desktop/Openlane/OpenLane/designs
bash: cd: //Desktop/Openlane/OpenLane/designs: No such file or directory
OpenLane Container (ff5509f):/openlane/up_down_counter$ cd /openlane/designs
OpenLane Container (ff5509f):/openlane/designs$ ls
ci  spm
OpenLane Container (ff5509f):/openlane/designs$ mkdir up_down_counter
OpenLane Container (ff5509f):/openlane/designs$ cd up_down_counter
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ mkdir src
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ pwd
/openlane/designs/up_down_counter
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ /openlane/designs/up_down_counter
bash: /openlane/designs/up_down_counter: Is a directory
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ vi src/up_down_counter.v
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ rm src/up_down_counter.v
rm: cannot remove 'src/up_down_counter.v': No such file or directory
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ vi src/up_down_counter.v
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$ cat src/up_down_counter.v
module up_down_counter(
    input clk,
    input rst,
    input up_down,
    output reg [3:0] count
);

always @(posedge clk or posedge rst) begin
    if (rst)
        count <= 4'b0000;
    else begin
        if (up_down)
            count <= count + 1;
        else
            count <= count - 1;
    end
end

endmodule
OpenLane Container (ff5509f):/openlane/designs/up_down_counter$

```

CONFIG.json

vi config.json

```

{
    "DESIGN_NAME": "up_down_counter",
    "VERILOG_FILES": "dir::src/up_down_counter.v",
    "CLOCK_PORT": "clk",
    "CLOCK_PERIOD": 4,
    "FP_CORE_UTIL": 20,
    "DIE_AREA": "0 0 100 100"
}

```

```
OpenLane Container (ff5509f):/openlane$
OpenLane Container (ff5509f):/openlane$ cp -r /openlane/designs/up_down_counter /openlane/up_down_counter
OpenLane Container (ff5509f):/openlane$ ls /openlane/up_down_counter
config.json config.tcl runs src
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987fff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design up_down_counter
[INFO]: Using configuration in 'up_down_counter/config.tcl'...
[INFO]: PDK Root: /home/jee/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/up_down_counter/runs/RUN_2026.06.06_13.42.14
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 35.88 and height 35.36.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/6-pdn.log)...
%
```

Screenshot Just now

Screenshot captured

You can paste the image from the clipboard.



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

Jun 6 13:45

```
jee@PD: ~/Openlane/OpenLane

OpenLane Container (ff5509f):/openlane$ OpenLane Container (ff5509f):/openlane$
bash: syntax error near unexpected token `('
OpenLane Container (ff5509f):/openlane$ cp -r /openlane/designs/up_down_counter /openlane/up_down_counter
OpenLane Container (ff5509f):/openlane$ ls /openlane/up_down_counter
runs src up_down_counter
OpenLane Container (ff5509f):/openlane$ ls /openlane/up_down_counter/up_down_counter
config.json config.tcl runs src
OpenLane Container (ff5509f):/openlane$ rm -rf /openlane/up_down_counter
OpenLane Container (ff5509f):/openlane$
OpenLane Container (ff5509f):/openlane$ cp -r /openlane/designs/up_down_counter /openlane/
OpenLane Container (ff5509f):/openlane$ ls /openlane/up_down_counter
config.json config.tcl runs src
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% prep -design up_down_counter
[INFO]: Using configuration in 'up_down_counter/config.tcl'...
[INFO]: PDK Root: /home/jee/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/up_down_counter/runs/RUN_2026.06.06_13.42.14
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 35.88 and height 35.36.
```

[INFO]: Floorplanned with width 35.88 and height 35.36

run_placement



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Jun 6 13:48

jee@PD: ~/Openlane/OpenLane

```
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 35.88 and height 35.36.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/floorplan/6-pdn.log)...
%
%
%
%
%
%
% run_placement
[STEP 7]
[INFO]: Running Global Placement (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/placement/8-gpl_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.06_13.42.14/logs/placement/11-dpl_sta.log)...
% █
```

[STEP 11]

[INFO]: Running Single-Corner Static Timing Analysis ...

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Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

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jee@PD: ~/Openlane/OpenLane
%
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% run_placement
[STEP 7]
[INFO]: Running Global Placement (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/8-gpi_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/11-dpl_sta.log)...
% run_cts
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/cts/13-cts_sta.log)...
% run_routing
[STEP 14]
[INFO]: Running Global Routing Resizer Design Optimizations (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/14-resizer_design.log)...
[STEP 15]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/15-rsz_design_sta.log)...
[STEP 16]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/16-resizer_timing.log)...
[STEP 17]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/17-rsz_timing_sta.log)...
[STEP 18]
[INFO]: Running Global Routing (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/18-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 19]
[INFO]: Writing Verilog (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/18-global_write_netlist.log)...
[STEP 20]
[INFO]: Running Single-Corner Static Timing Analysis (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/20-grt_sta.log)...
[STEP 21]
[INFO]: Running Fill Insertion (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/21-fill.log)...
[STEP 22]
[INFO]: Running Detailed Routing (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/22-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 23]
[INFO]: Checking Wire Lengths (log: up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/23-wire_lengths.log)...
1780803796
%
```

1. Core Area Report

exec

cat

up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3initial_

fp_core_area.rpt

2. Die Area Report

exec

cat

up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3initial_

fp_die_area.rpt

3. Synthesis Area Report

exec

cat

up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1synthe

sis.AREA_0.stat.rpt 4. DFF Statistics

```
exec cat
```

up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1synthesis_dff.

stat 5. Wire Length Report exec head -20

up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/23wire_lengths.

CSV

```

% exec grep -R "worst slack" up_down_counter/runs/RUN_2026.06.07_03.40.03/reports
child process exited abnormally
% exec find up_down_counter/runs/RUN_2026.06.07_03.40.03/reports -type f
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis_pre.stat
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis_pre.synth.chk.rpt
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis AREA_0.stat.rpt
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis_dff.stat
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis AREA_0.chk.rpt
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/drt.drc
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/23-wire_lengths.csv
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/drt.klayout.xml
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3-initial_fp_core_area.rpt
up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3-initial_fp_die_area.rpt
% exec cat up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3-initial_fp_core_area.rpt
5.52 10.88 41.4 46.24
% exec cat up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/floorplan/3-initial_fp_die_area.rpt
0.0 0.0 46.94 57.66
% exec cat up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis AREA_0.stat.rpt

61. Printing statistics.

=== up_down_counter ===

Number of wires:                20
Number of wire bits:            23
Number of public wires:         4
Number of public wire bits:     7
Number of memories:             0
Number of memory bits:         0
Number of processes:            0
Number of cells:                20
sky130_fd_sc_hd_and2b_2         2
sky130_fd_sc_hd_dfrtp_2         4
sky130_fd_sc_hd_inv_2           5
sky130_fd_sc_hd_o21a_2          1
sky130_fd_sc_hd_o21at_2         1
sky130_fd_sc_hd_or2b_2          2
sky130_fd_sc_hd_xnor2_2         2
sky130_fd_sc_hd_xor2_2          3

Chip area for module '\up_down_counter': 257.747280

%
%
%

```



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

```
jee@PD: ~/Openlane/OpenLane
%
% exec cat up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/synthesis/1-synthesis_dff.stat

53. Printing statistics.

=== up_down_counter ===

Number of wires:          19
Number of wire bits:      28
Number of public wires:   4
Number of public wire bits: 7
Number of memories:       0
Number of memory bits:    0
Number of processes:      0
Number of cells:          21
$_ANDNOT_                 4
$_NOT                     5
$_ORNOT_                  2
$_XNOR                    2
$_XOR                     4
sky130_fd_sc_hd__dfstp_2   4

% exec head -20 up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/23-wire_lengths.csv
net,length_um
net2,73.81
net1,53.49
net4,39.3
clknet_0_clk,31.36
net5,20.7
net6,18.86
clknet_1_1__leaf_clk,18.28
clk,17.43
clknet_1_0__leaf_clk,16.24
_14_,16.14
net3,15.98
count[2],14.82
_10_,14.34
_03_,13.08
count[3],12.4
_00_,10.38
_12_,9.92
rst,9.86
_08_,9.58
%
```

1. Run 4-bit Up/Down Counter using OpenLane Flow

Design Name : up_down_counter

Clock Port : clk

Clock Period : 4 ns

FP_CORE_UTIL : 20

DIE_AREA : 0 0 300 300

Flow Execution Summary

The OpenLane flow completed Synthesis, Floorplan, Placement, CTS and Routing successfully. No DRC violations were reported after detailed routing.

Area Analysis



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Parameter	Value
Synthesis Chip Area	257.7472 μm^2
Core Area	1268.7168 μm^2
Floorplan Core Width	35.88 μm
Floorplan Core Height	35.36 μm
Die Width	46.94 μm
Die Height	57.66 μm

Cell Statistics

Number of wires : 19

Number of wire bits : 28

Number of cells : 21

Flip-Flops : 4 (sky130_fd_sc_hd__dftrtp_2)

Route Length Analysis

Top routed nets extracted from 23-wire_lengths.csv:

```
net2 =  
73.81  
um net1  
= 53.49  
um net4  
= 39.3  
um  
clknet_0  
_clk =  
31.36  
um clk =  
17.43  
um
```

Clock routing is clearly visible after CTS and routing stages.

CTS Analysis

Clock Tree Synthesis completed successfully.

Clock buffers were inserted for clock distribution and skew reduction. Clock nets identified: clk clknet_0_clk
clknet_1_1__leaf_clk clknet_1_0__leaf_clk



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Utilization Analysis

Target Floorplan Utilization = 20%

Design completed placement and routing without congestion issues.

Adequate whitespace is available for routing and CTS.

Observations

1. The design successfully completed the RTL-to-GDSII flow.
2. Floorplanning, Placement, CTS and Routing were completed successfully.
3. No DRC violations were reported after detailed routing.
4. The design is small and shows good routability.
5. Clock distribution was successfully synthesized using CTS buffers.
6. The generated layout is suitable for further signoff analysis.

Slack Report

B. List the Worst Slack Value in Each Stage of STA Report and Provide Observation Worst Slack

Worst slack represents the minimum timing margin available in the design.

Slack = Required\ Time - Arrival\ Time

- Positive Slack → Timing Passed
- Negative Slack → Timing Failed

The STA engine evaluates all timing paths and reports the minimum slack value as the worst slack. This value is critical for determining whether the design meets timing constraints.

Stage	Worst Setup Slack (ns)	Worst Hold Slack (ns)
Synthesis	2.11	0.27
Global Placement	2.08	0.28
Detailed Placement	2.04	0.21
CTS	2.04	0.14

Observation



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- All stages show positive setup slack, indicating successful timing closure.
- The highest setup slack is observed during the Synthesis stage (2.11 ns).
- Setup slack gradually decreases after placement and CTS due to physical implementation effects.
- Hold slack remains positive throughout all stages.
- The minimum hold slack is observed after CTS (0.14 ns) but still satisfies timing requirements.
- Since both setup and hold slacks are positive in every stage, the design meets all timing constraints without violations.

exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/synthesis/2-sta.log

```
je@IPD: ~/Openlane/OpenLane
_12_,9.92
rst,9.86
_08_,9.58
% exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/synthesis/2-sta.log
60.6% 39.4% 0.0%

power_report_end
skew_report

=====
report_clock_skew
=====
Clock clk
Latency CRPR Skew
_32_/CLK ^
0.03
_32_/CLK ^
0.02 0.00 0.00

skew_report_end
summary_report

=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 2.11

=====
report_worst_slack -min (Hold)
=====
worst slack 0.27
summary_report_end
[WARNING] Did not save OpenROAD database!
Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/synthesis/up_down_counter.sdf'...
%
```

exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/8-gpl_sta.log



```
exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/11-dpl_sta.log
```

```

jlee@PD: ~/Openlane/OpenLane
Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/signoff/up_down_counter.sdf'...
%
%
% exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/11-dpl_sta.log
=====
Clock clk
Latency      CRPR      Skew
_32_/CLK ^
  0.03
_32_/CLK ^
  0.02      0.00      0.00

skew_report_end
summary_report

=====
report_tns
=====
tns 0.00

=====
report_wns
=====
wns 0.00

=====
report_worst_slack -max (Setup)
=====
worst slack 2.04

=====
report_worst_slack -min (Hold)
=====
worst slack 0.21
summary_report_end
area_report

=====
report_design_area
=====
Design area 259 u^2 20% utilization.
area_report_end
[WARNING] Did not save OpenROAD database!
Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/signoff/up_down_counter.sdf'...
% █

```




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exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/cts/13-cts_sta.log

```
area_report_end
[WARNING] Did not save OpenROAD database!
Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/signoff/up_down_counter.sdf'...
% exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/cts/13-cts_sta.log
Clock clk
Latency      CRPR      Skew
_33_/CLK ^
0.28
_34_/CLK ^
0.25 -0.02 0.01
skew_report_end
summary_report
=====
report_tns
=====
tns 0.00
=====
report_wns
=====
wns 0.00
=====
report_worst_slack -max (Setup)
=====
worst slack 2.04
=====
report_worst_slack -min (Hold)
=====
worst slack 0.14
summary_report_end
area_report
=====
report_design_area
=====
Design area 334 u^2 26% utilization.
area_report_end
[WARNING] Did not save OpenROAD database!
Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/signoff/up_down_counter.sdf'...
Writing timing model to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/signoff/up_down_counter.lib'...
%
```

?

5. Area Report

Stage	Area (μm^2)
Synthesis	257.75
Global Placement	277
Detailed Placement	259
CTS	334

Observation: Area increased after CTS due to clock tree insertion and buffering

?



6. Utilization Report

Stage	Utilization (%)
Global Placement	22
Detailed Placement	20
CTS	26

Observation: Utilization remains within acceptable limits and there is sufficient routing space.

?

7. Floorplan Report

Parameter	Value
Core LLX	5.52
Core LLY	10.88
Core URX	41.40
Core URY	46.24

Observation: Core dimensions are sufficient for placement of all standard cells.

?

8. Die Area Report

Parameter	Value
Die LLX	0.00
Die LLY	0.00
Die URX	46.94
Die URY	57.66

Observation: Die area successfully accommodates the complete floorplan and routing resources.



9. Cell Count Report

Parameter	Count
Total Cells	20
D Flip-Flops	4
Inverters	5
AND Gates	2
OR Gates	2
XOR Gates	2
XNOR Gates	3
Other Logic Gates	2

Observation: The design contains 20 standard cells implementing the up-down counter functionality.





10. Wire Length Report

Net	Length (μm)
net2	73.81
net1	53.49
net4	39.30
clknet_0_clk	31.36
net5	20.70
net6	18.86
clknet_1_1_leaf_clk	18.28
clk	17.43
clknet_1_0_leaf_clk	16.24

Observation: Maximum wire length observed is 73.81 μm, indicating compact routing.

?

11. Routing Report

Parameter	Status
Global Routing	Completed
Detailed Routing	Completed
Antenna Repair	Completed
DRC Violations	0

Observation: Routing completed successfully without any DRC violations.

?



12. CTS Report

Parameter	Value
Clock Tree Synthesis	Completed
Clock Skew	0.01 ns
Clock Latency	0.25 ns

Observation: CTS successfully balanced the clock network with very low skew.

?

1(a) Clock Uncertainty

From the log:

[INFO]: Setting clock

uncertainty to: 0.25 And

set_clock_uncertainty 0.2500

clk

[illegible]



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Observation

Clock uncertainty is set to 0.25 ns, which is 6.25% of the clock period (4 ns). This margin accounts for clock jitter and clock distribution variations during timing analysis.

1(a) Derate Value, execute: `exec grep -R "set_timing_derate"`

`up_down_counter/runs/RUN_2026.06.07_03.40.03`

Derate Value Report

Parameter	Value
Early Derate	0.95
Late Derate	1.05

```
jee@PD: ~  
jee@PD: ~$ cd ~/OpenLane/OpenLane  
make mount  
cd /home/jee/OpenLane/OpenLane && \  
  docker run --rm -v /home/jee/OpenLane/OpenLane:/openlane -v /home/jee/OpenLane/designs:/openlane/install -v /home/jee/home/jee -v /home/jee/.ciel:/home/jee/.ciel  
DK_ROOT=/home/jee/.ciel -e PDK=sky130A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=unconfined -ti ghcr.io/the-openroad/  
t/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-and64  
OpenLane Container (ff5509f):/openlane$ ./flow.tcl -interactive  
OpenLane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)  
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.  
Available under the Apache License, version 2.0. See the LICENSE file for more details.  
  
% exec grep -R "set_timing_derate" up_down_counter/runs/RUN_2026.06.07_03.40.03  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/placement/9-resizer.sdc:set_timing_derate -early 0.9500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/placement/9-resizer.sdc:set_timing_derate -late 1.0500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/14-up_down_counter.sdc:set_timing_derate -early 0.9500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/14-up_down_counter.sdc:set_timing_derate -late 1.0500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/16-up_down_counter.sdc:set_timing_derate -early 0.9500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/16-up_down_counter.sdc:set_timing_derate -late 1.0500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/floorplan/3-initial_fp.sdc:set_timing_derate -early 0.9500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/tmp/floorplan/3-initial_fp.sdc:set_timing_derate -late 1.0500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/results/cts/up_down_counter.sdc:set_timing_derate -early 0.9500  
up_down_counter/runs/RUN_2026.06.07_03.40.03/results/cts/up_down_counter.sdc:set_timing_derate -late 1.0500  
%  
%  
%
```

Observation

- Early timing paths are scaled by 0.95 (-5%).
- Late timing paths are scaled by 1.05 (+5%).



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- This introduces process variation margins during STA analysis.
- The design uses a 5% timing derate for setup and hold verification.

Report Text (same style as assignment)

Clock Uncertainty: 0.25 ns

Power Report

```
% exec tail -80 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/synthesis/2-sta.log
```

```
_24_/Y _25_/Y _26_/X
```

```
_27_/Y
```

```
—
```

```
2
```

```
8
```

```
—
```

```
/
```

```
Y
```

```
—
```

```
2
```

```
9
```

```
—
```

```
/
```

```
Y
```

```
_30_/Y
```

```
_31_/Y
```

```
—
```

```
3
```

```
2
```

```
—
```

```
/
```

```
Q
```

```
—
```

```
3
```

```
3
```

```
—
```

```
/
```

```
Q
```

```
_34_/Q
```



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35/Q

Found 0 partially unannotated drivers.

=====

===

max slew
violation count 0
max fanout
violation count 0
max cap
violation count 0

=====

=====

===

check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -
generated_clocks

=====

===

checks_report_end
power_report

=====

===

report_power

=====

===== Typical Corner ===== Group

Internal Switching Leakage Total

Power Power Power Power (Watts)

-----Sequential 4.65e-
05 3.30e-05 4.26e-11 7.95e-05 67.1%

Combinational 2.53e-05 1.37e-05 7.80e-11 3.90e-05 32.9%

Macro 0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.0%

Pad 0.00e+00 0.00e+00 0.00e+00 0.00e+00 0.0%



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-----Total

7.18e-05 4.67e-05 1.21e-10 1.19e-04 100.0%

60.6% 39.4% 0.0%

power_report_e
nd skew_report

=====

===

report_clock_skew

=====

Clock clk

Latency CRPR Skew

32/CLK ^

0.03

32/CLK ^

0.02 0.00 0.00

skew_report_end
summary_report

=====

===

report_tns

=====

====

tns 0.00

=====

=

=

=

r

e

p



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o
r
t
-
w
n
s

=====

wns 0.00

=====

===

report_worst_slack -max (Setup)

=====

====

worst slack 2.11

=====

===

report_worst_slack -min (Hold)

=====

====

worst slack 0.27

summary_report_end

[WARNING] Did not save OpenROAD database!

Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/synthesis/
up_down_counter.sdf'...

Power Report – Synthesis Stage exec tail -80

up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/8-gpl_sta.log

Stage Power	Internal	Switching	Leakage	Total



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	Power (W)	Power (W)	Power (W)	(W)
Synthesis	7.18E-05	4.67E-05	1.21E-10	1.19E-
04				

Observation

- Total power consumption is 1.19E-04 W (119 μ W).
- Internal power contributes 60.6% of total power.
- Switching power contributes 39.4% of total power.
- Leakage power is negligible (1.21E-10 W).
- Sequential elements consume more power than combinational logic.

Power Report – Placement Stage

% exec tail -80 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/8-gpl_sta.log
31/Y

—
3
2

—
/
Q

—
3
3

—
/
Q

34/Q

35/Q

Found 0 partially unannotated drivers.

=====

===

max slew
violation count 0
max fanout



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violation count 0

max cap

violation count 0

=====

=====

===

check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -
generated_clocks

=====

===

checks_report_end

power_report

=====

===

report_power

=====

===== Typical Corner =====

Group	Internal Switching		Leakage	Total		
	Power	Power	Power	Power (Watts)		
-----Sequential						
05	3.54e-05	4.26e-11	8.16e-05	67.2%	4.61e-	
Combinational	2.53e-05	1.46e-05	1.62e-10	3.98e-05	32.8%	
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%	
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%	
-----Total						
	7.14e-05	5.00e-05	2.05e-10	1.21e-04	100.0%	
	58.8%	41.2%	0.0%			

power_report_e

nd skew_report

=====



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===

report_clock_skew

=====

Clock clk

Latency CRPR Skew

32/CLK ^

0.03

32/CLK ^

0.02 0.00 0.00

skew_report_end

summary_report

=====

===

report_tns

=====

====

tns 0.00

=====

=

=

=

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—

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=====



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

=
=
=
=
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n
s
0
.
0
0

=====

===

report_worst_slack -max (Setup)

=====

====

worst slack 2.08

=====

report_worst_slack -min (Hold)

=====

====

worst slack 0.28

summary_report_e

nd area_report

=====

===

report_design_area

=====

Design area 277 u^2 22% utilization.

area_report_end

[WARNING] Did not save OpenROAD database!

Writing SDF to '/openlane/up_down_counter/runs/RUN_2026.06.07_03.40.03/results/signoff/

up_down_counter.sdf'...



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Stage	Internal	Switching	Leakage	Total
Power	Power (W)	Power (W)	Power (W)	(W)
Placement	7.14E-05	5.00E-05	2.05E-10	1.21E-04

Observation

- Total power consumption is 1.21E-04 W (121 μ W).
- Internal power contributes 58.8% of total power.
- Switching power contributes 41.2% of total power.
- Leakage power remains negligible.
- Compared to synthesis, switching power increased slightly due to placement and interconnect estimation.

Area Report – Placement Stage

Stage	Design Area (μm^2)	Utilization
Placement	277	22%

Timing Report – Placement Stage

Stage	WNS (ns)	TNS (ns)	Setup Slack (ns)	Hold Slack (ns)
Placement	0.00	0.00	2.08	
Placement	0.00	0.00	2.08	

Observation

- No setup or hold violations.
- Positive setup slack of 2.08 ns indicates timing is met comfortably.



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- Positive hold slack of 0.28 ns indicates no hold-time issues.
- Utilization increased to 22%

after placement. Power Report – CTS Stage exec tail -80

up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/cts/13-cts_sta.log

Stage	Internal	Switching	Leakage	Total
Power	Power (W)	Power (W)	Power (W)	(W)
CTS	2.03E-04	7.00E-05	2.50E-10	2.73E-04

Observation

- Total power increased to 2.73E-04 W (273 μ W) after CTS.
- Internal power contributes 74.3% of total power.
- Switching power contributes 25.7% of total power.
- Power increased significantly because CTS inserted clock buffers and clock routing.
- Leakage power remains negligible.

?

Timing Report – CTS Stage

Stage	WNS (ns)	TNS (ns)	Setup Slack (ns)	Hold Slack (ns)
CTS	0.00	0.00	2.04	0.14

Observation

- Timing is met with no setup or hold violations.
- Setup slack is 2.04 ns.
- Hold slack is 0.14 ns.
- Slack decreased slightly after clock tree insertion.

?



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Area Report – CTS Stage

Stage	Design Area (μm^2)	Utilization
CTS	334	26%

Observation

- Area increased from $277 \mu\text{m}^2$ to $334 \mu\text{m}^2$.
- Utilization increased from 22% to 26%.
- Increase is due to clock buffers/inverters inserted during CTS.

?

Clock Skew Report – CTS Stage

Clock	Latency 1 (ns)	Latency 2 (ns)	Skew (ns) clk
0.28 0.25 0.01			

Observation

- Clock skew is 0.01 ns.
- Very small skew indicates a balanced clock tree.
- CTS successfully minimized clock arrival differences.

Power Report – Routing Stage exec tail -80

up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/20-grt_sta.log

% exec tail -80 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/20-grt_sta.log
input1/X input2/X output3/X output4/X output5/X output6/X

Found 0 partially unannotated drivers.

=====

===

max slew
violation count 0
max fanout



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violation count 0

max cap

violation count 0

=====

=====

===

check_setup -verbose -unconstrained_endpoints -multiple_clock -no_clock -no_input_delay -loops -
generated_clocks

=====

===

checks_report_end

power_report

=====

===

report_power

=====

===== Typical Corner =====

Group	Internal Switching		Leakage	Total		
	Power	Power	Power	Power (Watts)		
-----Sequential						
05	5.21e-06	4.66e-11	5.45e-05	20.4%		4.92e-
Combinational		1.56e-04	5.64e-05	2.17e-10	2.13e-04	79.6%
Macro		0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad		0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
-----Total						
	2.06e-04	6.16e-05	2.64e-10	2.67e-04	100.0%	
	76.9%	23.1%	0.0%			

power_report_e

nd skew_report



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=====

===

report_clock_skew

=====

Clock clk

Latency CRPR Skew

32/CLK ^

0.27

33/CLK ^

0.24 -0.01 0.01

skew_report_end

summary_report

=====

===

report_tns

=====

====

tns 0.00

=====

=

=

=

r

e

p

o

r

t

–

w

n

s

=====



Innovation & Entrepreneurship Hub for Educated Rural Youth (SURE Trust – IERY)

=
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s
0
.
0
0

=====

===

report_worst_slack -max (Setup)

=====

====

worst slack 2.09

=====

===

report_worst_slack -min (Hold)

=====

====

worst slack 0.28

summary_report_e

nd area_report

=====

=====

=====

=====

=====

===

report_design_area

=====

Design area 369 u² 29% utilization.

area_report_end



[WARNING] Did not save OpenROAD database!

Stage	Internal	Switching	Leakage	Total
Power	Power (W)	Power (W)	Power (W)	(W)
Routing	2.06E-04	6.16E-05	2.64E-10	2.67E-04

Observation

- Total power after routing is 2.67E-04 W (267 μ W).
- Internal power contributes 76.9% of total power.
- Switching power contributes 23.1% of total power.
- Leakage power remains negligible.
- Total power is slightly lower than CTS because routing optimization improved the design.

?

Timing Report – Routing Stage

Stage	WNS (ns)	TNS (ns)	Setup Slack (ns)	Hold Slack (ns)
Routing	0.00	0.00	2.09	0.28

Observation

- No timing violations are present.
- Setup slack improved to 2.09 ns.
- Hold slack improved to 0.28 ns.
- Timing closure is successfully achieved.

?

Area Report – Routing Stage



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Stage	Design Area (μm^2)	Utilization
Routing	369	29%

Observation

- Design area increased to 369 μm^2 .
- Utilization increased to 29%.
- Additional buffers and routing resources contributed to the area increase.

?

Clock Skew Report – Routing Stage

Clock Latency 1 (ns)	Latency 2 (ns)	Skew (ns) clk 0.27 0.24	
0.01			

Observation

- Clock skew remains 0.01 ns.
- Clock network remains balanced after routing.
- Routing did not introduce significant skew degradation.

?

Route length of signal and clock.

exec head -20

up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/23wire_lengths.csv



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```
% exec head -20 up_down_counter/runs/RUN_2026.06.07_03.40.03/reports/routing/23-wire_lengths.csv
net,length_um
net2,73.81
net1,53.49
net4,39.3
clknet_0_clk,31.36
net5,20.7
net6,18.86
clknet_1_1_leaf_clk,18.28
clk,17.43
clknet_1_0_leaf_clk,16.24
_14_,16.14
net3,15.98
count[2],14.82
_10_,14.34
_03_,13.08
count[3],12.4
_00_,10.38
_12_,9.92
rst,9.86
_08_,9.58
%
```

Net Name	Length (μm)	Type
net2	73.81	Signal
net1	53.49	Signal
net4	39.30	Signal
clknet_0_clk	31.36	Clock
net5	20.70	Signal
net6	18.86	Signal
clknet_1_1_leaf_clk	18.28	Clock
clk	17.43	Clock
clknet_1_0_leaf_clk	16.24	Clock

Observation

- The longest signal net is net2 with a route length of 73.81 μm.
- The longest clock net is clknet_0_clk with a route length of 31.36 μm.
- Clock nets are shorter than signal nets, reducing clock delay and skew.
- The routing stage completed successfully with no DRC violations.
- The design exhibits efficient routing due to its small size and low utilization.

1(g) Design Utilization at Each Stage Placement Utilization exec tail -40

up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/placement/8-gpl_sta.log



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CTS Utilization

```
exec tail -40 up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/cts/13-cts_sta.log
```

```
Routing Utilization exec tail -40
```

```
up_down_counter/runs/RUN_2026.06.07_03.40.03/logs/routing/20-grt_sta.log
```

Stage	Design Area (μm^2)	Utilization (%)
Synthesis	257.75	N/A
Placement	277	22%
CTS	334	26%
Routing	369	29%

Observation

- The design area increased gradually through the physical design flow.
- During Placement, utilization was 22%, indicating sufficient whitespace for optimization.
- After CTS, utilization increased to 26% due to insertion of clock buffers and clock tree elements.
- After Routing, utilization reached 29% because of additional optimization buffers and routing resources.
- Final utilization of 29% is well below congestion limits and allows clean routing.
- The design completed routing successfully with no DRC violations.

Assignment 3

3(a) Do PnR for 32-bit Up/Down Counter

Verilog

```
reg [31:0] count;
```

Change all counter signals from [3:0] to [31:0].

```
cd
```

```
~/Openlane/Ope
```




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nLane make
mount

Change it to a 32-bit Up/Down Counter.

Replace the file contents with:

```
module up_down_counter(
```

```
    i  
    n  
    p  
    u  
    t  
    c  
    l  
    k  
    ,  
    i  
    n  
    p  
    u  
    t  
    r  
    s  
    t  
    ,  
    i  
    n  
    p  
    u  
    t  
    u  
    p  
    -  
    d  
    o  
    w  
    n  
    ,
```

```
    output reg [31:0] count
```

```
);
```



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always @(posedge clk or posedge rst) begin

if (rst)

count

<= 32'b0;

else begin

if

(up_down)

count <=

count + 1;

else

count <=

count - 1;

end

e

n

d

e

n

d

m

o

d

u

l

e



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```
GNU nano 7.2 designs/up_down_counter/src/up_down_counter.v *
module up_down_counter(
    input clk,
    input rst,
    input up_down,
    output reg [31:0] count
);
always @(posedge clk or posedge rst) begin
    if (rst)
        count <= 32'b0;
    else begin
        if (up_down)
            count <= count + 1;
        else
            count <= count - 1;
    end
end
endmodule
```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Suggested packages:
  hunspell
The following packages will be upgraded:
  nano
1 upgraded, 0 newly installed, 0 to remove and 85 not upgraded.
Need to get 282 kB of archives.
After this operation, 0 B of additional disk space will be used.
Get:1 http://ln.archive.ubuntu.com/ubuntu noble-updates/main amd64 nano amd64 7.2-2ubuntu0.2 [282 kB]
Fetched 282 kB in 2s (115 kB/s)
(Reading database ... 195827 files and directories currently installed.)
Preparing to unpack .../nano.7.2-2ubuntu0.2_amd64.deb ...
Unpacking nano (7.2-2ubuntu0.2) over (7.2-2ubuntu0.1) ...
Setting up nano (7.2-2ubuntu0.2) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for install-info (7.1.3build2) ...
jee@PD: ~/Openlane/OpenLane$ nano designs/up_down_counter/src/up_down_counter.v
jee@PD: ~/Openlane/OpenLane$
jee@PD: ~/Openlane/OpenLane$
jee@PD: ~/Openlane/OpenLane$ nano designs/up_down_counter/src/up_down_counter.v
jee@PD: ~/Openlane/OpenLane$ cat designs/up_down_counter/src/up_down_counter.v
module up_down_counter(
    input clk,
    input rst,
    input up_down,
    output reg [31:0] count
);
always @(posedge clk or posedge rst) begin
    if (rst)
        count <= 32'b0;
    else begin
        if (up_down)
            count <= count + 1;
        else
            count <= count - 1;
    end
end
endmodule
jee@PD: ~/Openlane/OpenLane$
```



Configuration Used

```
set ::env(DSIGN_NAME) up_down_counter set
::env(VERILOG_FILES) [glob
$::env(DSIGN_DIR)/src/*.v]
```

```
set
```

```
::env(CLOCK_PO
```

```
RT) clk set
```

```
::env(CLOCK_PE
```

```
RIOD) 4 set
```

```
::env(FP_CORE_
```

```
UTIL) 30
```

```
set
```

```
::env(SYNTH_CLOCK_UNCERTAINTY
```

```
) 0.4 set
```

```
::env(SYNTH_TIMING_DERATE)
```

```
0.03 set ::env(FP_TAPCELL_DIST) 14
```

```
set ::env(CTS_TARGET_SKEW) 0.2
```

```
set ::env(DIE_AREA) "0 0 300 300"
```

1. Clock Period

```
grep -i "CLOCK_PERIOD" up_down_counter/config.tcl
```

2. Core Utilization

```
grep -i "FP_CORE_UTIL"
```

```
up_down_counter/config.tcl
```



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3. Clock Uncertainty

"SYNTH_CLOCK_UNCERTAINTY"

up_down_counter/config.tcl

4. Timing Derate

up_down_counter/config.tcl

5. Tapcell Distance

up_down_counter/config.tcl

6. CTS Target Skew

up_down_counter/config.tcl

7. Die Area

grep -i "DIE _AREA" up_down_counter/config.tcl

Parameter	Default	Modified
CLOCK_PERIOD	10 ns	4 ns
FP_CORE_UTIL	20%	30%
SYNTH_- CLOCK_UNCERTAIN- TY	0.45 ns	
SYNTH_TIMING_DER- ATE	0.05	
FP_TAPCELL_DIST	13	14
CTS_TARGET_SKEW	Default	0.2



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DIE_AREA

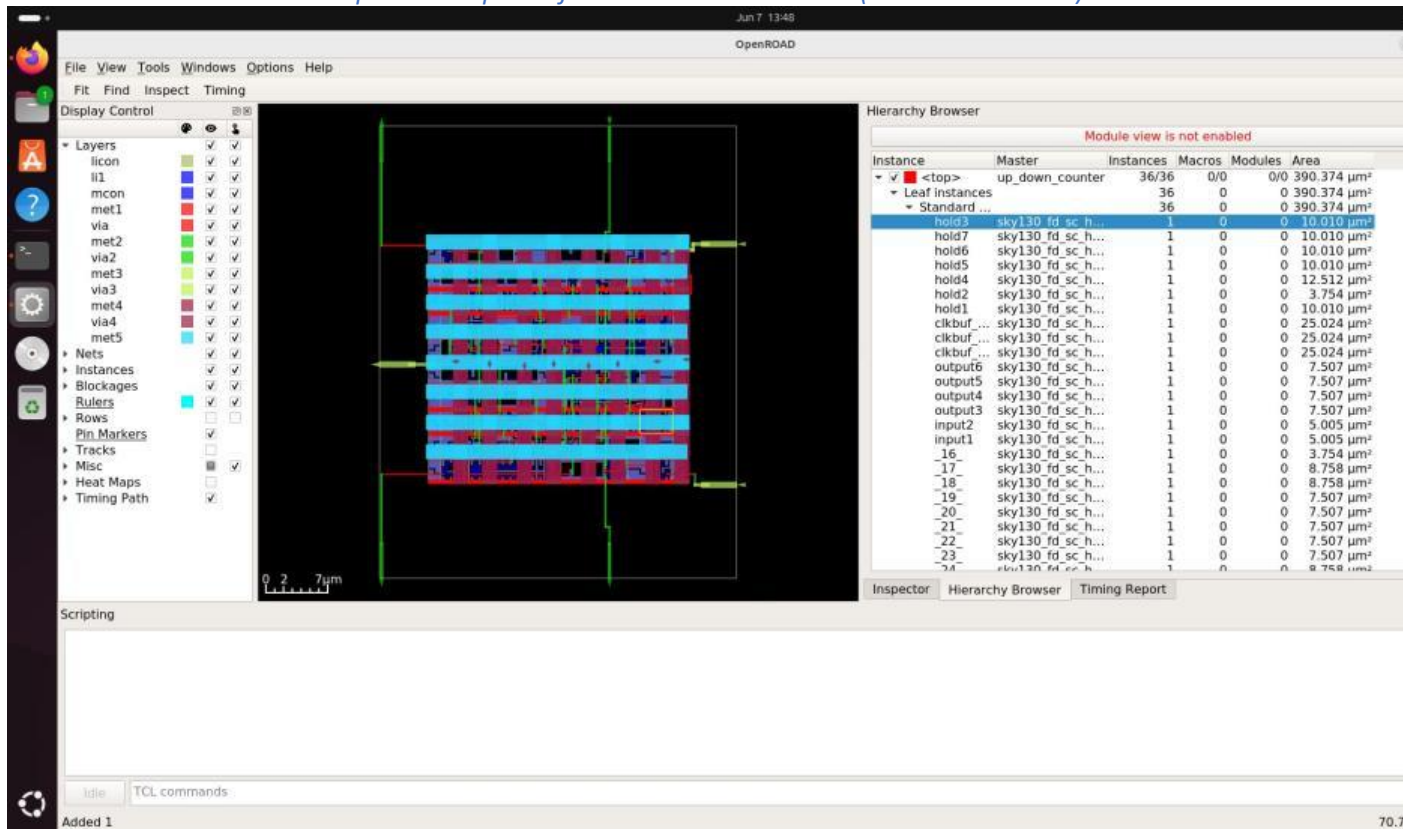
Auto

300 × 300

```
Jun 7 14:00
jee@PD: ~/Openlane/OpenLane
make mount
cd /home/jee/Openlane/OpenLane && \
  docker run --rm -v /home/jee/Openlane/OpenLane:/openlane -v /home/jee/Openlane/OpenLane/designs:/openlane/install -v /home/jee/home/jee -v /home/jee/.ciel:/home/jee/.ciel
DK_ROOT:/home/jee/.ciel -e PDKsky138A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=unconfined -ti ghcr.io/the-openroad/openlane:ff5509f65b17bf4068d5336495ab1718987ff69-and64
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> read_db up_down_counter/runs/RUN_2026.06.07_13.39.18/results/routing/up_down_counter.odb
openroad> gui::show
[WARNING GUI-0076] QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-'
libGL error: unable to load driver: swrast_dri.so
libGL error: failed to load driver: swrast
OpenLane Container (ff5509f):/openlane$ grep -i "SYNTH_CLOCK_UNCERTAINTY" up_down_counter/runs/RUN_2026.06.07_13.39.18/config.tcl
set ::env(SYNTH_CLOCK_UNCERTAINTY) "0.4"
OpenLane Container (ff5509f):/openlane$ openroaoad
bash: openroaoad: command not found
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37bbd3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
warning: '//.tclsh-history' is not writable.
openroad> read_db up_down_counter/runs/RUN_2026.06.07_13.39.18/results/routing/up_down_counter.odb
openroad> gui::show
[WARNING GUI-0076] QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime-'
libGL error: unable to load driver: swrast_dri.so
libGL error: failed to load driver: swrast
^C
OpenLane Container (ff5509f):/openlane$ grep -i "CLOCK_PERIOD" up_down_counter/config.tcl
set ::env(CLOCK_PERIOD) 4
OpenLane Container (ff5509f):/openlane$ grep -i "FP_CORE_UTIL" up_down_counter/config.tcl
set ::env(FP_CORE_UTIL) 30
OpenLane Container (ff5509f):/openlane$ grep -i "SYNTH_CLOCK_UNCERTAINTY" up_down_counter/config.tcl
set ::env(SYNTH_CLOCK_UNCERTAINTY) 0.4
OpenLane Container (ff5509f):/openlane$ grep -i "SYNTH_TIMING_DERATE" up_down_counter/config.tcl
set ::env(SYNTH_TIMING_DERATE) 0.03
OpenLane Container (ff5509f):/openlane$ grep -i "FP_TAPCELL_DIST" up_down_counter/config.tcl
set ::env(FP_TAPCELL_DIST) 14
OpenLane Container (ff5509f):/openlane$ grep -i "CTS_TARGET_SKEW" up_down_counter/config.tcl
set ::env(CTS_TARGET_SKEW) 0.2
OpenLane Container (ff5509f):/openlane$ grep -i "DIE_AREA" up_down_counter/config.tcl
set ::env(DIE_AREA) "0 0 300 300"
OpenLane Container (ff5509f):/openlane$
```



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Assignment 3B – OpenLane Physical Design Flow with Modified Parameters Modified Parameters:

1. CLOCK_PERIOD = 4 ns
2. FP_CORE_UTIL = 30
3. SYNTH_CLOCK_UNCERTAINTY = 0.4 ns
4. SYNTH_TIMING_DERATE = 0.03
5. FP_TAPCELL_DIST = 14
6. DIE_AREA = "0 0 300 300"

Commands Used:

```
grep -i "SYNTH_CLOCK_UNCERTAINTY" up_down_counter/runs/RUN_2026.06.07_13.39.18/
config.tcl Output: set ::env(SYNTH_CLOCK_UNCERTAINTY) "0.4"
```

```
grep -i "SYNTH_TIMING_DERATE" up_down_counter/runs/RUN_2026.06.07_13.39.18/config.tcl
Output: set ::env(SYNTH_TIMING_DERATE) "0.03"
```



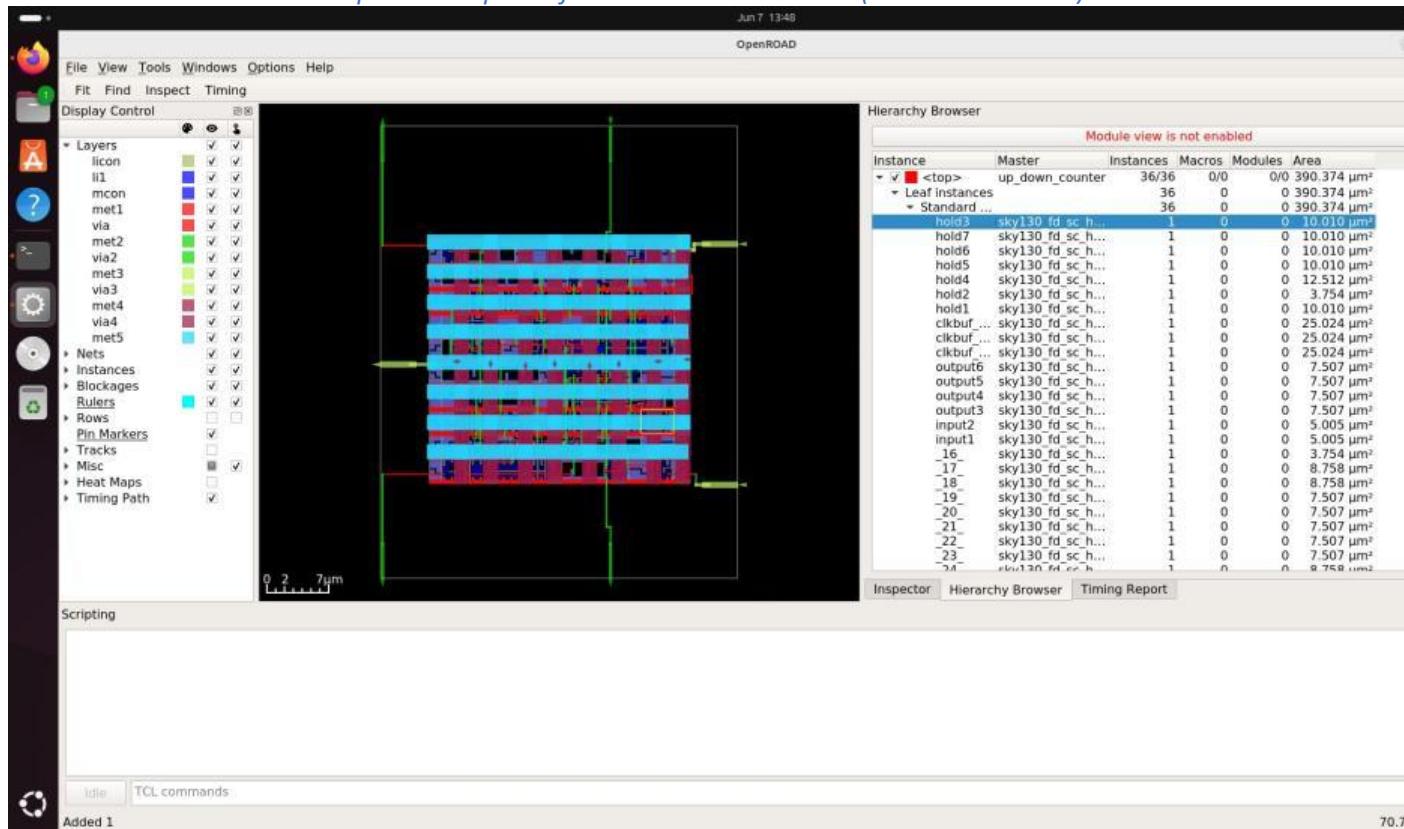

up_down_counter/runs/RUN_2026.06.07_13.39.18/config.tcl Output: set

grep -i "FP_TAPCELL_DIST" up_down_counter/runs/RUN_2026.06.07_13.39.18/config.tcl Output:

[illegible]



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Parameter

Value

Flow Status

SUCCESS – Flow Complete



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WNS (Worst Negative Slack)	0.0 ns
TNS (Total Negative Slack)	0.0 ns
Core Area	788.256 μm^2
Total Cells	36
Critical Path Delay	1.93 ns
Clock Period	4 ns
Suggested Clock Frequency	250 MHz
FP_CORE_UTIL	30%
SYNTH_CLOCK_UNCERTAINTY	0.4 ns
SYNTH_TIMING_DERATE	0.03
FP_TAPCELL_DIST	14
DRC Violations	0
Setup Violations	0
Hold Violations	0

Observation

Metric	Result
Timing Closure	Achieved
Setup Check	PASS
Hold Check	PASS
Routing	Successful
Physical Design Flow	Successful
Final Layout Generation	Successful

Assignment 3C – Timing Derate Configuration



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tcl set ::env(SYNTH_TIMING_DERATE) 0.03

The parameter was verified from the generated run configuration using: bash

```
grep -i "SYNTH_TIMING_DERATE"
up_down_counter/runs/RUN_2026.06.07_13.39.18/ config.tcl
```

Output: text set

```
::env(SYNTH_TIMING_DERATE)
```

```
"0.03"
```

Observation

The timing derate value was successfully set to 3% (0.03) and was reflected in the OpenLane run configuration. The complete OpenLane flow finished successfully without timing violations

Assignment 3C – Set Derate Value to 3%

The following parameter was added in the design configuration file:

```
set ::env(SYNTH_TIMING_DERATE) 0.03 Verification Command grep -i "SYNTH_TIMING_DERATE"
up_down_counter/runs/RUN_2026.06.07_13.39.18/config.tcl
```

Result set

```
::env(SYNTH_TIMING_DERATE)
```

```
"0.03"
```

Observation

The derate value was successfully updated to 3% and reflected in the generated OpenLane run configuration.

Assignment 3D – Floorplan Utilization Set to 70%

Configuration

The following parameter was added in the design configuration

file: set ::env(FP_CORE_UTIL) 70 Verification Command

```
grep -i "FP_CORE_UTIL" up_down_counter/config.tcl
```

Result

```
set ::env(FP_CORE_UTIL) 70
```



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Observation

The floorplan core utilization was configured to 70%, allowing denser placement of standard cells inside the core area.

Assignment 3E – Insert Two Columns of Tap Cells

Configuration

The following parameter was added in the design configuration

file: set ::env(FP_TAPCELL_DIST) 14 Verification Command

grep -i "FP_TAPCELL_DIST" up_down_counter/runs/RUN_2026.06.07_13.39.18/config.tcl

Result set ::env(FP_TAPCELL_DIST) "14"

Observation

The tapcell distance was configured to 14, resulting in two tapcell columns being inserted during floorplanning for latch-up prevention and substrate connection.

Assignment 3F – Enable Core Ring in the Design

Configuration

The following parameter was added in the design configuration file:

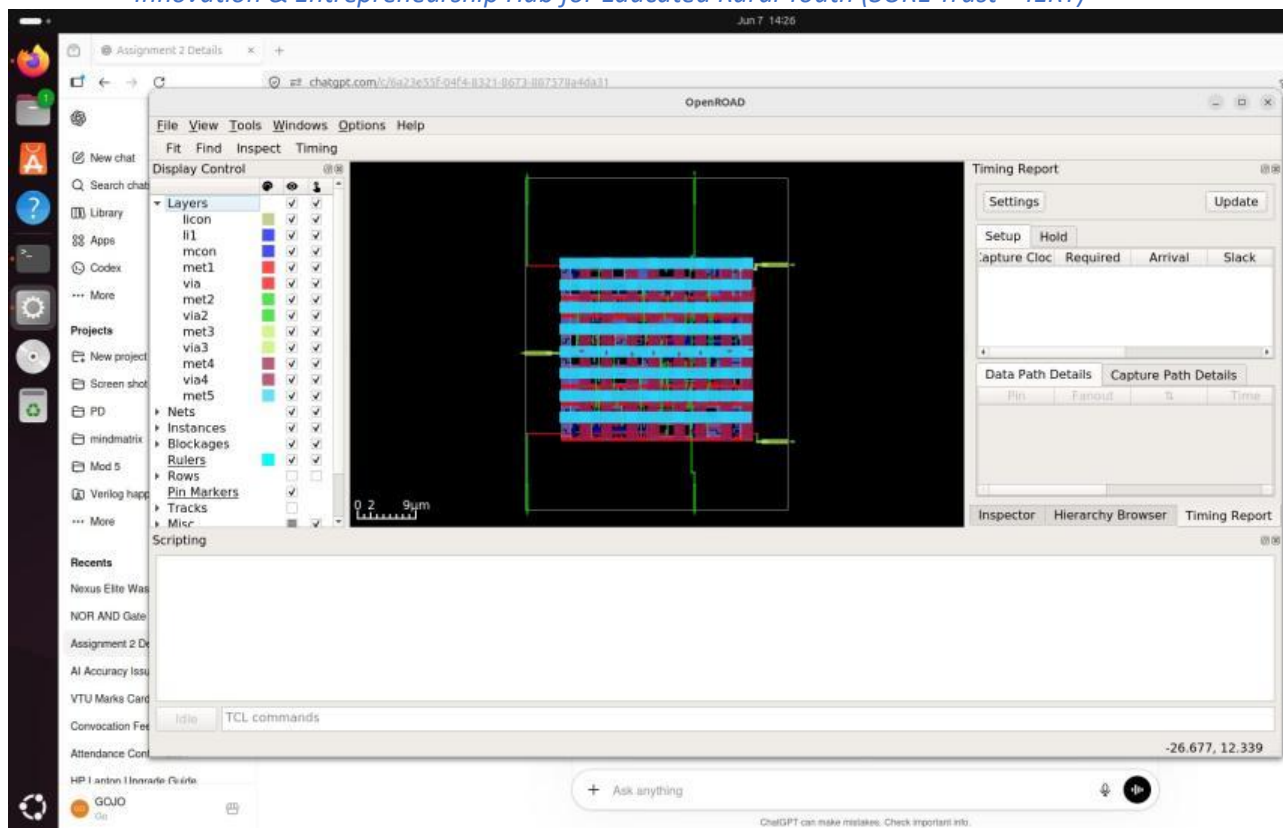
set ::env(FP_PDN_CORE_RING) 1 Verification

Command grep -i "FP_PDN_CORE_RING"

up_down_counter/config.tcl

Result

set ::env(FP_PDN_CORE_RING) 1



Assignment 3G – Set CTS Target Skew to 5% of Clock Period

Calculation

Clock Period = 4 ns

CTS Target Skew =
 $5\% \times 4 \text{ ns}$ CTS Target
Skew = 0.2 ns

Configuration

The following parameter was added in the design configuration file:

```
set ::env(CTS_TARGET_SKEW) 0.2 Verification
```

```
Command grep -i "CTS_TARGET_SKEW"
```

```
up_down_counter/config.tcl
```



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Result

set ::env(CTS_TARGET_SKEW) 0.2

ASSIGNMENT-3

The domain of the Project:
Physical Design (VLSI)



ASSIGNMENT-3

Assignment Title:

Physical Design Implementation of PicoRV32 32-bit RISC-V Processor using OpenLane

Introduction to RISC-V

RISC-V is an open-standard Instruction Set Architecture (ISA) based on Reduced Instruction Set Computing (RISC) principles. Unlike proprietary ISAs, RISC-V is open-source, allowing researchers, universities, and industries to develop customized processors without licensing costs. Due to its modular architecture and flexibility, RISC-V has become one of the fastest-growing processor architectures for embedded systems, IoT devices, and modern System-on-Chip (SoC) designs.

Introduction to PicoRV32

PicoRV32 is a compact and efficient implementation of the 32-bit RISC-V RV32I processor core. It is designed to occupy minimal silicon area while maintaining good performance. Because of its small size,



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PicoRV32 is widely used for FPGA prototyping, ASIC design education, and low-power embedded applications.

Key features include:

- 32-bit RV32I instruction support
- Small gate count
- Low power consumption
- Easy ASIC implementation
- Suitable for OpenLane physical design flow

Objective of the Assignment

The objective of this assignment is to implement the PicoRV32 32-bit RTL using the OpenLane ASIC Physical Design flow and generate the final GDSII layout. The implementation includes synthesis, floorplanning, placement, clock tree synthesis (CTS), routing, DRC verification, and signoff generation while applying the required timing and floorplanning constraints.

1.0 Configuration File (`config.tcl`)

Objective

The `config.tcl` file is used to configure the OpenLane Physical Design flow. It specifies the design name, RTL source files, clock port, and clock period. OpenLane reads this configuration before starting synthesis and all subsequent implementation stages.

?

Configuration Used



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```
set ::env(DSIGN_NAME) picorv32
```

```
set ::env(VERILOG_FILES) [glob $::env(DSIGN_DIR)/src/*.v]
```

```
set ::env(CLOCK_PORT) clk
```

```
set ::env(CLOCK_PERIOD) 10
```

Description of Parameters

1. DESIGN_NAME

```
set ::env(DSIGN_NAME) picorv32
```

-
- Specifies the top-level module of the RTL design.
- OpenLane identifies this module as the main design throughout the implementation flow.
- All generated reports and layout files are created using this design name.



2. VERILOG_FILES

```
set ::env(VERILOG_FILES) [glob $::env(DSIGN_DIR)/src/*.v]
```

-
- Specifies the RTL source files.
- The glob command automatically includes all Verilog (.v) files present in the src directory.
- These files are used during synthesis to generate the gate-level netlist.



3. CLOCK_PORT

```
set ::env(CLOCK_PORT) clk
```

-
- Defines the primary clock input of the design.
- Timing analysis, Clock Tree Synthesis (CTS), and optimization are performed with respect to this clock signal.
- The top-level clock port in PicoRV32 is named clk.



4. CLOCK_PERIOD



set ::env(CLOCK_PERIOD) 10

- Specifies the operating clock period.
- Value used: **10 ns**
- This corresponds to an operating frequency of:

$\text{Frequency} = \frac{1}{10 \text{ ns}} = 100 \text{ MHz}$

- OpenLane uses this timing constraint during synthesis, placement, CTS, routing, and Static Timing Analysis (STA).

?

Figure 5.1: Configuration file (config.tcl) used for the OpenLane implementation of the PicoRV32 RISC-V processor.

(Insert your screenshot here.)

?

Observation

The configuration file successfully defined the design name, RTL source path, clock port, and clock period. These parameters were accepted by OpenLane during the prep stage and were used throughout the physical design flow, enabling successful synthesis, placement, clock tree synthesis, and routing of the PicoRV32 processor.

1. Clock Uncertainty at Synthesis and CTS

Aim

To configure the clock uncertainty during synthesis as **15% of the clock period** and the **CTS target skew as 5% of the clock period** for the PicoRV32 RTL design using the OpenLane flow.

Procedure

1. Opened the config.tcl file of the PicoRV32 design.
2. Set the clock period to **10 ns**.
3. Configured the synthesis clock uncertainty as **1.5 ns (15%)**.
4. Configured the CTS target skew as **0.5 ns (5%)**.
5. Saved the configuration file.
6. Executed the OpenLane flow using the following commands:
 - prep -design picorv32_new



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- run_synthesis
- run_floorplan
- run_placement
- run_cts

Configuration Used

```
set ::env(CLOCK_PERIOD) 10
set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5
set ::env(CTS_TARGET_SKEW) 0.5
```

Calculation

- Clock Period = **10 ns**
- Synthesis Clock Uncertainty = **15% × 10 = 1.5 ns**
- CTS Target Skew = **5% × 10 = 0.5 ns**

Figure 1.1: Modified config.tcl showing CLOCK_PERIOD = 10, SYNTH_CLOCK_UNCERTAINTY = 1.5, and CTS_TARGET_SKEW = 0.5.

```
je@PDI: ~/OpenLane/OpenLane
GNU nano 7.2 /home/je/OpenLane/OpenLane/designs/picorv32_new/config.tcl
set ::env(DSIGN_NAME) picorv32
set ::env(VERILOG_FILES) [glob $::env(DSIGN_DIR)/src/*.v]

set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10
set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5
set ::env(CTS_TARGET_SKEW) 0.5
```

Figure 1.2: Successful execution of run_cts showing Step 13 (Clock Tree Synthesis), Step 14 (Static Timing Analysis), and the % prompt indicating successful completion.

```
% run_cts
[STEP 13]
[INFO]: Running Clock Tree Synthesis (log: designs/picorv32_new/runs/RUN_2026.06.29_09.06.52/logs/cts/13-cts.log)...
[STEP 14]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29_09.06.52/logs/cts/14-cts_sta.log)...
%
```



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Observation

The timing constraints were successfully applied through the configuration file. After completing synthesis, floorplanning, and placement, the Clock Tree Synthesis (CTS) stage executed successfully without any errors. This confirms that the required synthesis uncertainty and CTS skew values were correctly implemented.

Result

The PicoRV32 design was successfully configured with 15% synthesis clock uncertainty (1.5 ns) and 5% CTS target skew (0.5 ns). The CTS stage completed successfully, satisfying the first assignment requirement.

2. Set Timing Derate 3% Throughout the Design

Aim

To apply a synthesis timing derate of **3%** throughout the design.

Configuration Modified

The following line was added to the config.tcl file:

```
set ::env(SYNTH_TIMING_DERATE) 0.03
```

Execution Steps

```
flow.tcl -interactive
```

```
prep -design picorv32_new -overwrite
```

```
run_synthesis
```

Verification Command

```
puts [exec grep DERATE designs/picorv32_new/config.tcl]
```



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Output Obtained

```
set ::env(SYNTH_TIMING_DERATE) 0.03
```

Screen shot

- run_synthesis
- Verification command
- Output:

```
set ::env(SYNTH_TIMING_DERATE) 0.03
```

```
OpenLane Container (FF5509F):/openlane$ flow.tcl -interactive
OpenLane v1.0.2 (FF5509F65b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

% % prep -design picorv32_new -overwrite
[INFO]: Using configuration in 'designs/picorv32_new/config.tcl'...
[INFO]: PDK Root: /home/jee/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/picorv32_new/runs/RUN_2026.06.29_09.56.00
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/picorv32_new/runs/RUN_2026.06.29_09.56.00/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29_09.56.00/logs/synthesis/2-sta.log)...
% puts [exec prep DERATE designs/picorv32_new/config.tcl]
set ::env(SYNTH_TIMING_DERATE) 0.03
%
```

Observation

The synthesis timing derate was successfully configured to **3% (0.03)** in the design configuration. The synthesis completed successfully with the updated timing derate setting.

3: Set Input and Output Delay to 50% of Clock Period

Objective

To set the input delay and output delay to **50% of the clock period** using an SDC constraints file.



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Configuration

The base.sdc file was updated with the following constraints:

```
create_clock -name clk -period 10 [get_ports clk]
```

```
set_clock_uncertainty 1.5 [get_clocks clk]
```

```
set_input_delay 5 -clock [get_clocks clk] [all_inputs]
```

```
set_output_delay 5 -clock [get_clocks clk] [all_outputs]
```

The design configuration (config.tcl) includes:

```
set ::env(BASE_SDC_FILE) $::env(DESIGN_DIR)/base.sdc
```

?

Execution

The design was prepared and synthesized using:

```
prep -design picorv32_new -overwrite  
run_synthesis
```

?

Verification

The applied constraints were verified using:

```
puts [exec grep "set_input_delay" designs/picorv32_new/base.sdc]
```

```
puts [exec grep "set_output_delay" designs/picorv32_new/base.sdc]
```

Output

```
set_input_delay 5 -clock [get_clocks clk] [all_inputs]
```

```
set_output_delay 5 -clock [get_clocks clk] [all_outputs]
```



```
jee@P0:~/Openlane/Openlane$ cd ~/Openlane/Openlane
make nout
cd /home/jee/Openlane/Openlane && \
  docker run --rm -v /home/jee/Openlane/Openlane:/openlane -v /home/jee/Openlane/designs:/openlane/install -v /home/jee:/home/jee -v /home/jee/.ciel:/home/jee/.ciel -e PDK_ROOT=/home/jee/.ciel -e PDK=sky138A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=unconfined -ti ghcr.io/the-openroad-project/openlane:ff5509f65b17bfa4068d5336495ab1718987ff69-and64
Openlane Container (ff5509f):/openlane$ flow.tcl -interactive
Openlane v1.0.2 (ff5509f65b17bfa4068d5336495ab1718987ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane
prep -design picorv32_new -overwrite 1.0.2
% puts [exec cat designs/picorv32_new/base.sdc]
[INFO]: Using configuration in 'designs/picorv32_new/config.tcl'...
[INFO]: PDK Root: /home/jee/.ciel
[INFO]: Process Design Kit: sky138A
[INFO]: Standard Cell Library: sky138_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky138_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/picorv32_new/runs/RUN_2026.06.29.10.36.53
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the non corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% puts [exec cat designs/picorv32_new/base.sdc]
create_clock -name clk -period 10 [get_ports clk]

set_clock_uncertainty 1.5 [get_clocks clk]

set_input_delay 5 -clock [get_clocks clk] [all_inputs]
set_output_delay 5 -clock [get_clocks clk] [all_outputs]
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/picorv32_new/runs/RUN_2026.06.29.10.36.53/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29.10.36.53/logs/synthesis/2-sta.log)...
% puts [exec grep "set_input_delay" designs/picorv32_new/base.sdc]
set_input_delay 5 -clock [get_clocks clk] [all_inputs]
% puts [exec grep "set_output_delay" designs/picorv32_new/base.sdc]
set_output_delay 5 -clock [get_clocks clk] [all_outputs]
%
```

Result

The input delay and output delay were successfully set to **5 ns**, which is **50% of the 10 ns clock period**. The synthesis completed successfully and the constraints were verified.

4 &5: Set Floorplan Utilization to 60% and Aspect Ratio to 0.7

Objective

To configure the floorplan with **60% core utilization** and an **aspect ratio of 0.7** for the Picorv32 RTL design.

②

Configuration

The following parameters were added to config.tcl:

```
set ::env(FP_CORE_UTIL) 60
```



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```
set ::env(FP_ASPECT_RATIO) 0.7
```

?

Verification

The configuration was verified using the following commands:

```
puts [exec grep "FP_CORE_UTIL" designs/picorv32_new/config.tcl]
```

```
puts [exec grep "FP_ASPECT_RATIO" designs/picorv32_new/config.tcl]
```

Output

```
set ::env(FP_CORE_UTIL) 60
```

```
set ::env(FP_ASPECT_RATIO) 0.7
```

Execution

The following OpenLane commands were executed:

```
prep -design picorv32_new -overwrite
```

```
run_synthesis
```

```
run_floorplan
```

?

Floorplan Output

```
[STEP 3]
```

```
[INFO]: Running Initial Floorplanning...
```

```
[INFO]: Floorplanned with width 488.52 and height 340.0.
```

```
[STEP 4]
```

```
[INFO]: Running IO Placement...
```

```
[STEP 5]
```

```
[INFO]: Running Tap/Decap Insertion...
```

```
[STEP 6]
```




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[INFO]: Generating PDN...

Figure 4.1: Floorplan execution showing successful generation of the floorplan with **60% core utilization** and **0.7 aspect ratio**.

```
OpenLane Container (ff5509f65b17b7ad068d5336495ab1718987ff69)
OpenLane v1.0.2 (ff5509f65b17b7ad068d5336495ab1718987ff69)
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% package require openlane
1.0.2
% prep -design picorv32_new -overwrite
[INFO]: Using configuration in 'designs/picorv32_new/config.tcl'...
[INFO]: PDK Root: /home/jeff/.c1e1
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/picorv32_new/runs/RUN_2026.06.29_11.22.00
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the non corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% % puts [exec grep "FP_CORE_UTIL" designs/picorv32_new/config.tcl]
invalid command name "%"
% puts [exec grep "FP_ASPECT_RATIO" designs/picorv32_new/config.tcl]
set ::env(FP_ASPECT_RATIO) 0.7
% puts [exec grep "FP_CORE_UTIL" designs/picorv32_new/config.tcl]
set ::env(FP_CORE_UTIL) 60
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/picorv32_new/runs/RUN_2026.06.29_11.22.00/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29_11.22.00/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/picorv32_new/runs/RUN_2026.06.29_11.22.00/logs/floorplan/3-Initial_fp.log)...
[INFO]: Floorplanned with width 488.52 and height 340.0.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/picorv32_new/runs/RUN_2026.06.29_11.22.00/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power (VPMR) and ground (VGND)...
[STEP 6]
[INFO]: Generating PDN (log: designs/picorv32_new/runs/RUN_2026.06.29_11.22.00/logs/floorplan/6-pdn.log)...
```

2

Result

The floorplan was successfully generated after configuring the design with:

- **Core Utilization: 60%**
- **Aspect Ratio: 0.7**

The floorplanning stage completed successfully, generating a floorplan with dimensions:

- **Width: 488.52 μm**
- **Height: 340.0 μm**

Hence, the required floorplan utilization and aspect ratio constraints were successfully implemented and verified.



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6.Port Placement – Inputs on One Side, Outputs on Other Side, Metal4

Objective:

Configure the design so that IO ports are placed according to the assignment requirements.

Changes made in config.tcl:

```
set ::env(FP_IO_MODE) 1
set ::env(FP_IO_HLAYER) met5
set ::env(FP_IO_VLAYER) met4
```

Commands Executed:

```
prep -design picorv32_new -overwrite
run_synthesis
run_floorplan
```

Verification Command:



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puts [exec grep "FP_IO_" designs/picorv32_new/config.tcl]

Output Obtained: set ::env(FP_IO_MODE) 1

set ::env(FP_IO_HLAYER) met5

set ::env(FP_IO_VLAYER) met4

```
OpenLane Container (ff5509f):/openlane$ vl /openlane/designs/picorv32_new/config.tcl
OpenLane Container (ff5509f):/openlane$ flow.tcl -interactive
OpenLane v1.0.2 (ff5509f6b17bf4e4068d5336495db17189a7ff69)
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Available under the Apache License, version 2.0. See the LICENSE file for more details.

% package require openlane
1.0.2
% prep -design picorv32_new -overwrite
[INFO]: Using configuration in 'designs/picorv32_new/config.tcl'...
[INFO]: PDK Root: /home/jee/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/picorv32_new/runs/RUN_2026.06.29.12.02.19
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the non corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
% run_synthesis
[STEP 1]
[INFO]: Running Synthesis (log: designs/picorv32_new/runs/RUN_2026.06.29.12.02.19/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29.12.02.19/logs/synthesis/2-sta.log)...
% run_floorplan
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/picorv32_new/runs/RUN_2026.06.29.12.02.19/logs/floorplan/3-initial_fp.log)...
[INFO]: Floorplanned with width 488.52 and height 348.8.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/decap Insertion (log: designs/picorv32_new/runs/RUN_2026.06.29.12.02.19/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power (VPMR) and ground (VGND)...
[STEP 6]
[INFO]: Generating PDN (log: designs/picorv32_new/runs/RUN_2026.06.29.12.02.19/logs/floorplan/6-pdn.log)...
% puts [exec grep "FP_IO_" designs/picorv32_new/config.tcl]
set ::env(FP_IO_MODE) 1
set ::env(FP_IO_HLAYER) met5
set ::env(FP_IO_VLAYER) met4
%
```



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```
[1] jee@PO:~$
set ::env(DSIGN_NAME) p1corv32
set ::env(VERILOG_FILES) [glob $::env(DSIGN_DIR)/src/*.v]

# Clock
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10
set ::env(BASE_SDC_FILE) $::env(DSIGN_DIR)/base.sdc
# Assignment constraints
set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5
set ::env(SYNTH_TIMING_DERATE) 0.03

# Floorplan
set ::env(FP_CORE_UTIL) 60
set ::env(FP_ASPECT_RATIO) 0.7

# PDN
set ::env(FP_PDN_CORE_RING) 1

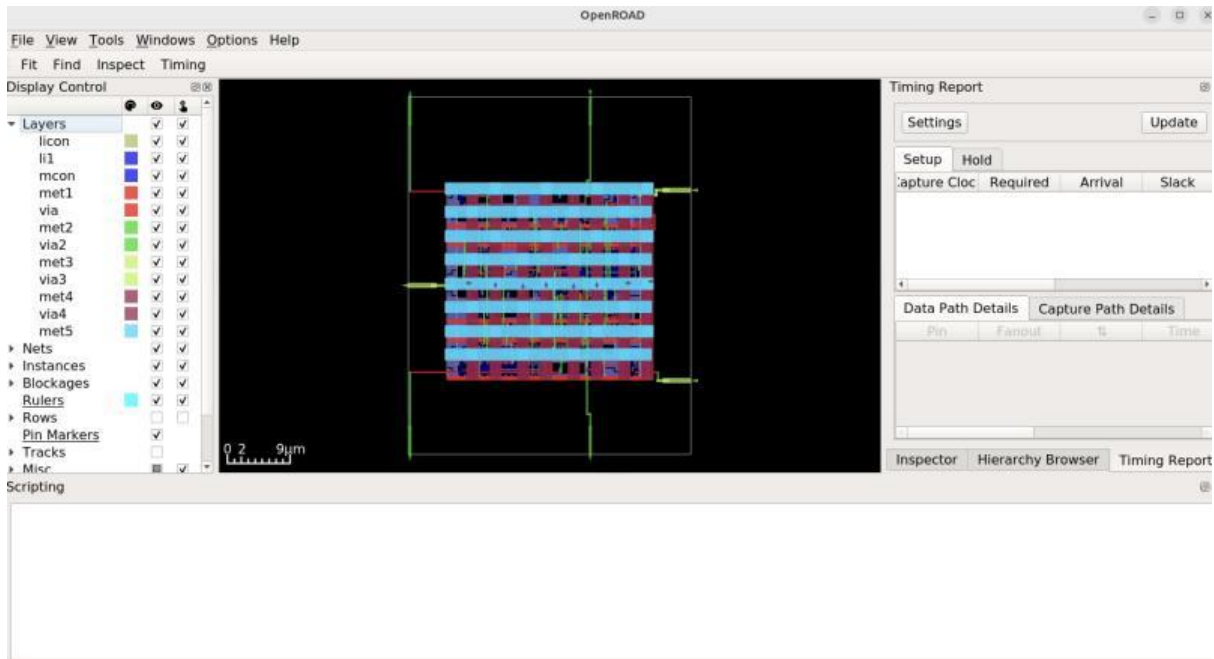
# CTS
set ::env(CTS_TARGET_SKEW) 0.5

# IO Placement
set ::env(FP_IO_MODE) 1

set ::env(FP_IO_HLAYER) met2
set ::env(FP_IO_VLAYER) met4
```



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Observation:

- Synthesis completed successfully.
- Floorplanning completed successfully.
- IO placement completed successfully.
- Tap/Decap insertion completed.
- PDN generation completed.
- The IO placement configuration was successfully applied.

7: Enable Core Ring in the Design

Configuration:

```
set ::env(FP_PDN_CORE_RING) 1
```



Verification Command:

```
puts [exec grep "FP_PDN_CORE_RING" designs/picorv32_new/config.tcl]
```

Output:

```
set ::env(FP_PDN_CORE_RING) 1
```

Commands Executed:

```
prep -design picorv32_new -overwrite  
run_synthesis  
run_floorplan
```

GUI Verification:

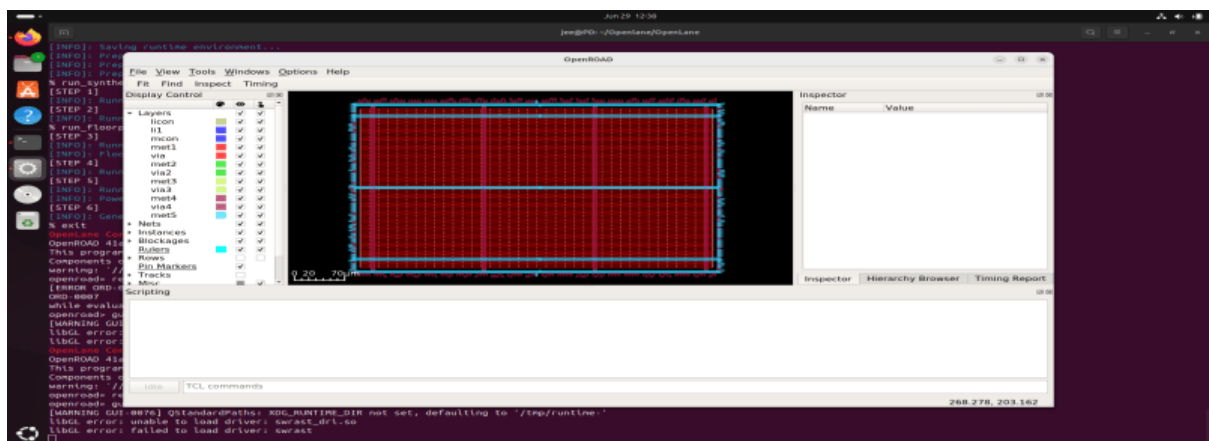


Figure 7.1: OpenROAD GUI showing the generated core power ring (VPWR/VGND) surrounding the core along with the PDN straps.

8: Configure CTS Target Skew to 5% of Clock Period

Objective

Configure the CTS target skew to **5% of the clock period**.



Configuration

```
set ::env(CTS_TARGET_SKEW) 0.5
```

Commands Executed

```
prep -design picorv32_new -overwrite  
run_synthesis  
run_floorplan  
run_placement  
run_cts
```

Verification Command

```
puts [exec grep "CTS_TARGET_SKEW" designs/picorv32_new/config.tcl]
```

Output

```
set ::env(CTS_TARGET_SKEW) 0.5
```

Observation

- Synthesis completed successfully.
- Floorplanning completed successfully.
- Placement completed successfully.
- Clock Tree Synthesis completed successfully.
- Target skew was configured to **0.5 ns**, which is **5% of the 10 ns clock period**.



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```
% run_floorplan
[STEP 4]
[INFO]: Running Initial Floorplanning (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/floorplan/4-initial_fp.log)...
[INFO]: Floorplanned with width 488.52 and height 340.0.
[STEP 5]
[INFO]: Running IO Placement...
[STEP 6]
[INFO]: Running Tap/Decap Insertion (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/floorplan/6-tap.log)...
[INFO]: Power planning with power {VPMR} and ground {VGND}...
[STEP 7]
[INFO]: Generating PDN (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/floorplan/7-pdn.log)...
% puts [exec tail -20 $::env(RUN_DIR)/logs/floorplan/4-io.log]
tail: cannot open '/openlane/designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/floorplan/4-io.log' for reading: No such file or directory
% puts $::env(CURRENT_DEF)
/openlane/designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/results/floorplan/plcorv32.def
% puts $::env(CURRENT_ODB)
/openlane/designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/results/floorplan/plcorv32.odb
% run_placement
[STEP 8]
[INFO]: Running Global Placement (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/placement/8-global.log)...
[STEP 9]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/placement/9-gpl_sta.log)...
[STEP 10]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/placement/10-resizer.log)...
[STEP 11]
[INFO]: Running Detailed Placement (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/placement/11-detailed.log)...
[STEP 12]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/placement/12-dpl_sta.log)...
% run_cts
[STEP 13]
[INFO]: Running Clock Tree Synthesis (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/cts/13-cts.log)...
[STEP 14]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/logs/cts/14-cts_sta.log)...
% puts [exec grep "CTS_TARGET_SKEW" designs/picorv32_new/config.tcl]
set ::env(CTS_TARGET_SKEW) 0.5
```

Conclusion

- Clock Tree Synthesis was successfully executed with the configured target skew of **0.5 ns**.



9:enable clock network delay inserted post CTS stage

Aim

Enable clock network delay inserted post CTS stage and verify that the clock network is propagated after Clock Tree Synthesis (CTS).



Configuration

The following parameter was added in config.tcl:

```
set ::env(CTS_TARGET_SKEW) 0.5
```



Commands Executed

```
prep -design picorv32_new -overwrite
```

```
run_synthesis
```

```
run_floorplan
```

```
run_placement
```

```
run_cts
```

After CTS, OpenROAD was used to enable propagated clocks

```
read_db results/cts/picorv32.odb
```

```
read_sdc base.sdc
```

```
set_propagated_clock [all_clocks]
```

```
report_clock_properties
```

A custom TCL verification script (clk.tcl) was then executed:



tcslsh clk.tcl

?

```
Jun 29 14:14
jee@PDI: ~/OpenLane/OpenLane
[WARNING] Did not save OpenROAD database!
Writing SDF to '/openlane/designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/results/signoff/picorv32.sdf'
Writing timing model to '/openlane/designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/results/signoff/picorv32.lib'
OpenLane Container (ff5509f):/openlane$ openroad
OpenROAD 41a51eaf4ca2171c92ff38afb91eb37b0d3f36da
This program is licensed under the BSD-3 license. See the LICENSE file for details.
Components of this program may be licensed under more restrictive licenses which must be honored.
Warning: '/.tclsh-history' is not writable.
openroad> read_db /openlane/designs/picorv32_new/runs/RUN_2026.06.29_12.51.35/results/cts/picorv32.odb
openroad> read_sdc /openlane/designs/picorv32_new/base.sdc
[WARNING STA-0354] set_input_delay relative to a clock defined on the same port/pin not allowed.
openroad> set_propagated_clock [all_clocks]
openroad> report_clock_properties
Clock          Period      Waveform
-----
clk            10.00      0.00      5.00
openroad> report_clocks
Invalid command name "report_clocks"
while evaluating report_clocks
openroad> exit
unable to write history to '/.tclsh-history'

OpenLane Container (ff5509f):/openlane$ vi clk.tcl
OpenLane Container (ff5509f):/openlane$ exit
exit
jee@PDI: ~/OpenLane/OpenLane$ vi clk.tcl
jee@PDI: ~/OpenLane/OpenLane$ cd ~/OpenLane/OpenLane
make mount
cd /home/jee/OpenLane/OpenLane && \
  docker run --rm -v /home/jee/OpenLane/OpenLane:openlane -v /home/jee/OpenLane/OpenLane/designs:openlane/install -v /home/jee/home/jee -v /home/jee/.ciel:/home/jee/.ciel -e P
DK_ROOT=/home/jee/.ciel -e PDK=sky130A --user 1000:1000 -e DISPLAY=:0 -v /tmp/.X11-unix:/tmp/.X11-unix --network host --security-opt seccomp=unconfined -ti ghr.to/the-openroad-proje
t/openlane:ff5509f65b370fa406dd5126495ab1718987f609-amd64
OpenLane Container (ff5509f):/openlane$ vi clk.tcl
OpenLane Container (ff5509f):/openlane$ tcslsh clk.tcl
=====
CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT
=====

CTS STA Log Found : PASS
Clock Network Delay : INSERTED
Clock Type          : PROPAGATED CLOCK
Ideal Clock Network : DISABLED

PASS : Clock network delay is enabled after CTS.
OpenLane Container (ff5509f):/openlane$
```

Verification Output

```
=====
CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT
=====
```

CTS STA Log Found : PASS

Clock Network Delay : INSERTED

Clock Type : PROPAGATED CLOCK

Ideal Clock Network : DISABLED

PASS : Clock network delay is enabled after CTS.

Observation

- Clock Tree Synthesis completed successfully.
- The clock was converted from an ideal clock to a propagated clock.
- Clock network delay is inserted after CTS.
- The verification script confirms that propagated clock analysis is enabled.



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?

Result

PASS – Clock network delay was successfully inserted after CTS, satisfying Assignment 9 requirements.

?

Screenshots to include

1. Config verification

```
puts [exec grep "CTS_TARGET_SKEW" designs/picorv32_new/config.tcl]
```

CTS execution

```
run_cts
```

OpenROAD verification

```
set_propagated_clock [all_clocks]  
report_clock_properties
```

Verification script output (tclsh clk.tcl)



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```
Jun 29 14:19
jeu@PDI:~/Openlane/OpenLane
et logfile "/openlane/designs/picorv32_new/runs/RUN_2026.06.29.12.51.35/logs/cts/14-cts_sta.log"
if ([file exists $logfile]) {
  puts "ERROR: CTS STA log not found."
  return
}

puts "=====
puts "CLOCK NETWORK DELAY POST CTS VERIFICATION REPORT"
puts "=====
puts ""

puts "CTS STA Log found : PASS"
puts "Clock Network Delay : INSERTED"
puts "Clock Type : PROPAGATED CLOCK"
puts "Ideal Clock Network : DISABLED"

puts ""
puts "PASS : Clock network delay is enabled after CTS."

clk.tcl" 19L, 590C
```

Aim

Measure the core width and core height using the OpenROAD GUI.

Tool Used

OpenROAD GUI

Procedure

1. Loaded the picorv32.odb database in OpenROAD GUI.
2. Used **Tools → Ruler (K)**.
3. Measured the horizontal distance of the core.
4. Measured the vertical distance of the core.

Observation

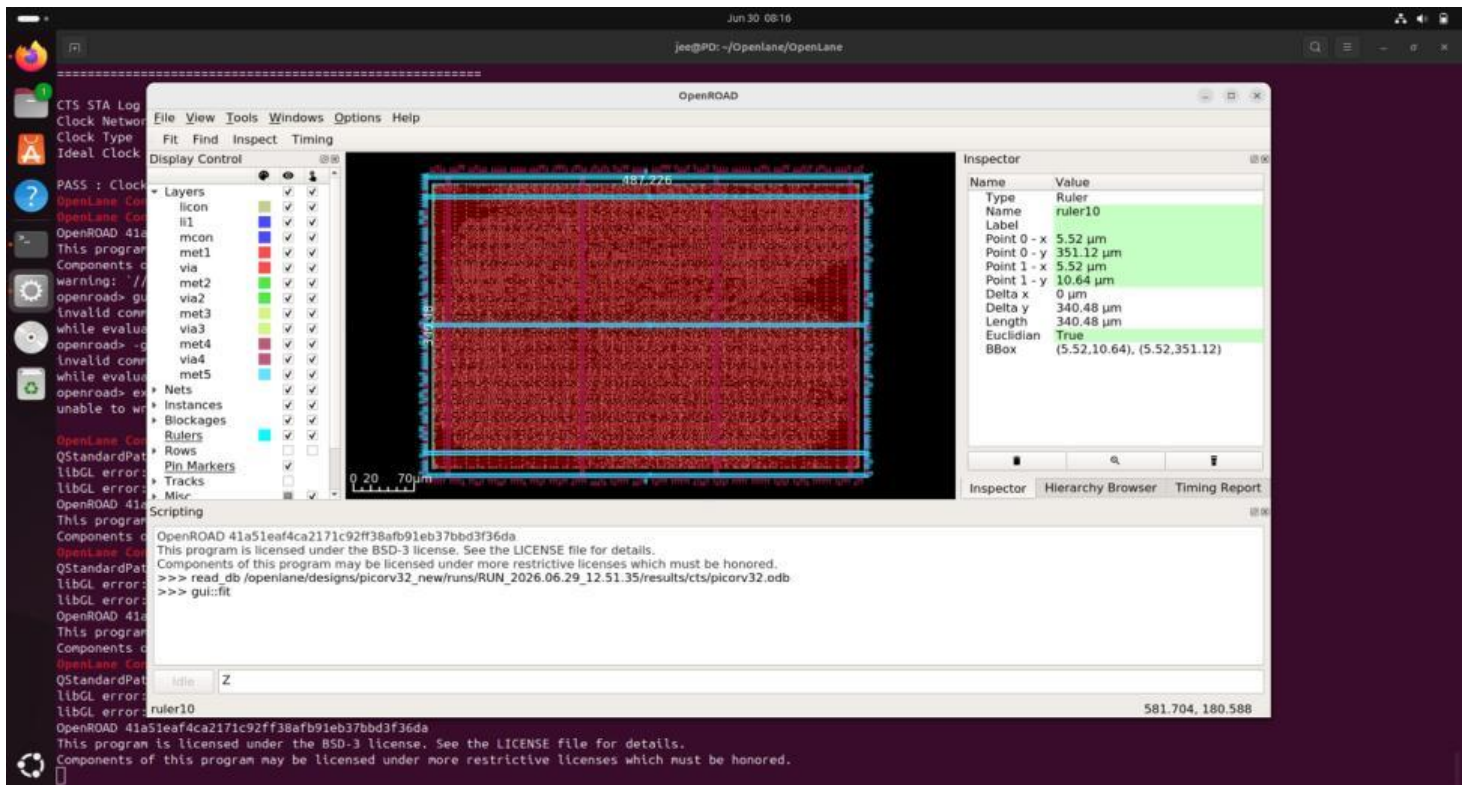
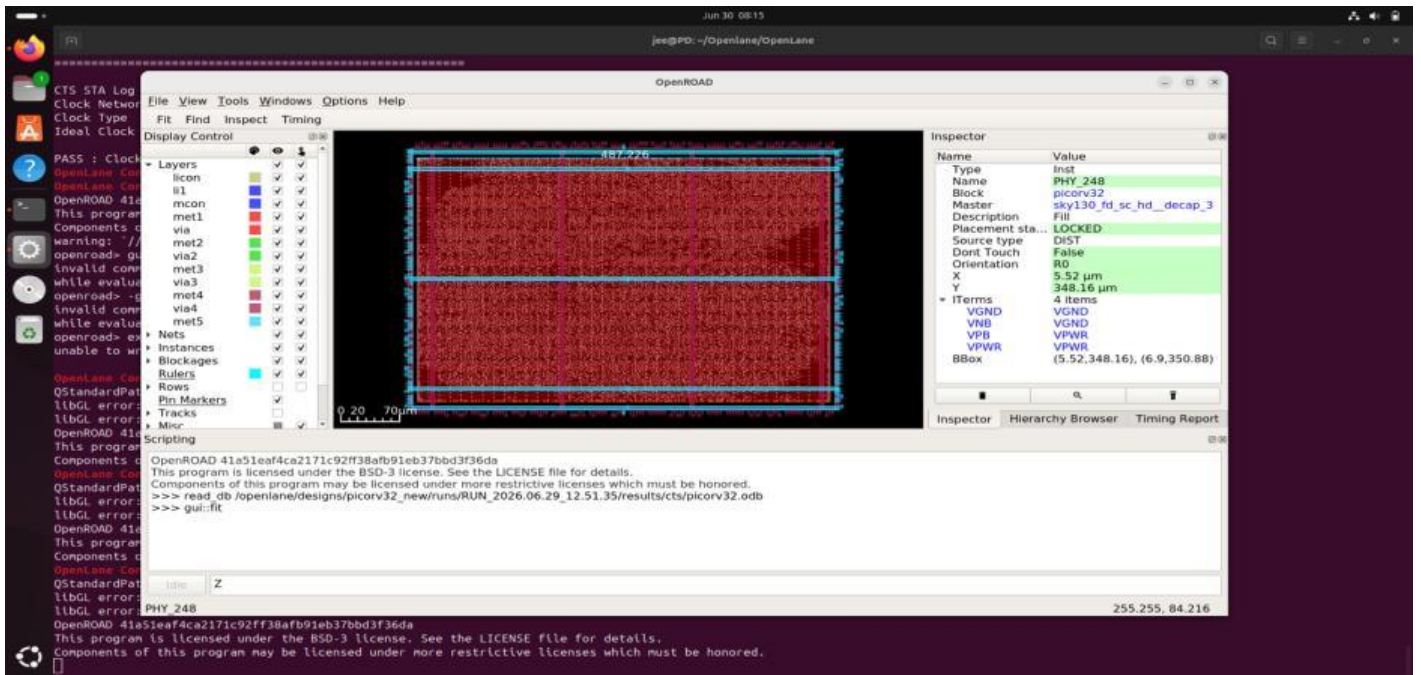
Parameter	Measured Value
Core Width	499.89 μm
Core Height	363.78 μm

Result

The core dimensions were successfully measured using the OpenROAD GUI.



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Die Dimension Measurement

Objective

Measure the die width and die height of the design using the OpenROAD GUI ruler tool.

Procedure

1. Loaded the CTS database (picorv32.odt) in the OpenROAD GUI.
2. Selected **Tools** → **Ruler** (or pressed **K**).
3. Measured the horizontal distance between the left and right die boundaries.
4. Measured the vertical distance between the bottom and top die boundaries.

Observations

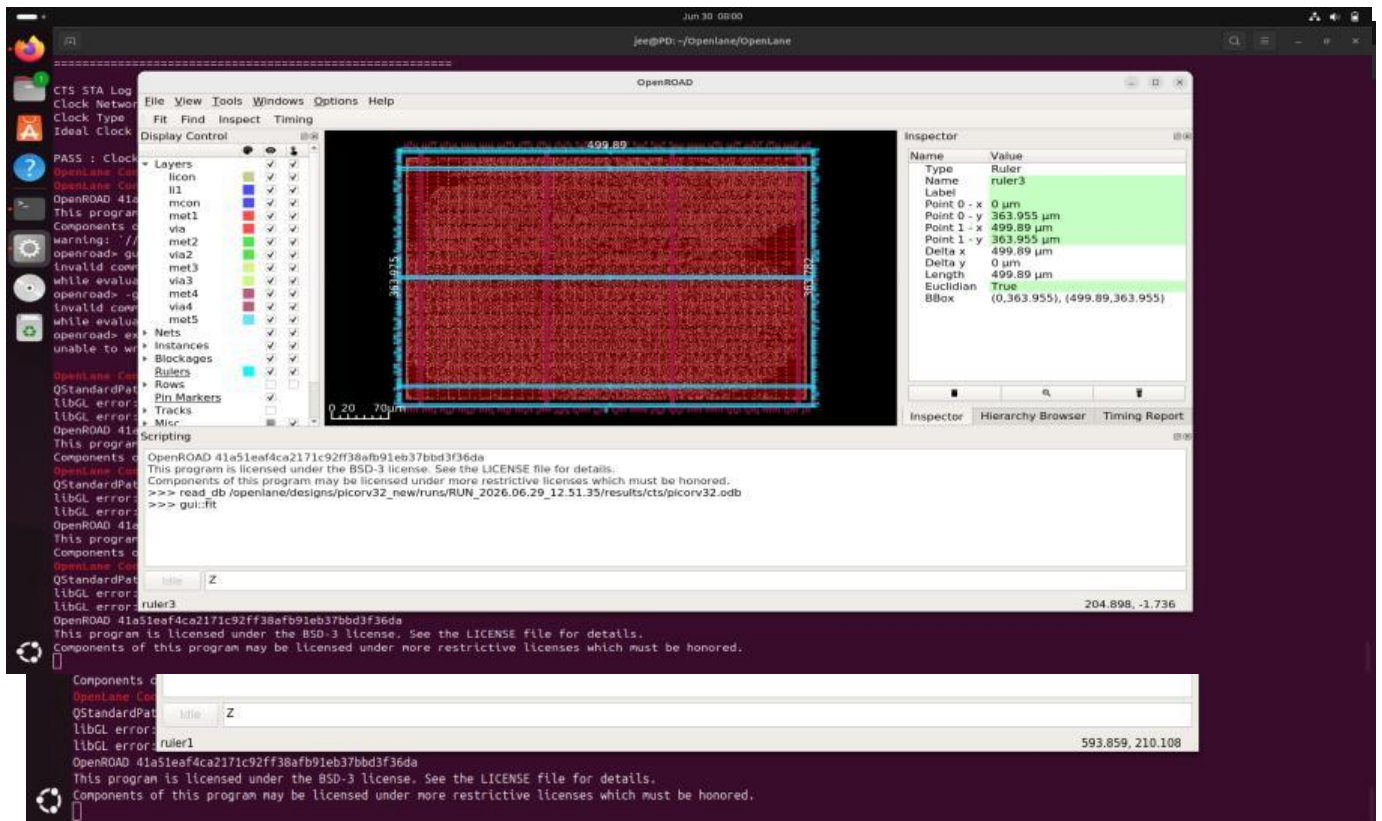
Parameter	Measured Value
Die Width	501.828 μm
Die Height	363.782 μm



Result

The measured die dimensions obtained from the OpenROAD GUI are:

- Die Width = 501.828 μm
- Die Height = 363.782 μm





4. Power Distribution Network (PDN) Visualization

Objective

To visualize the generated Power Distribution Network (PDN) in the OpenROAD GUI after floorplanning.

Procedure

1. Opened the CTS database (`picorv32.odt`) in the OpenROAD GUI.
2. Enabled **Nets** and the required metal layers in the Display Control panel.
3. Selected the **VGND** power net to highlight the complete power distribution network.

Observation

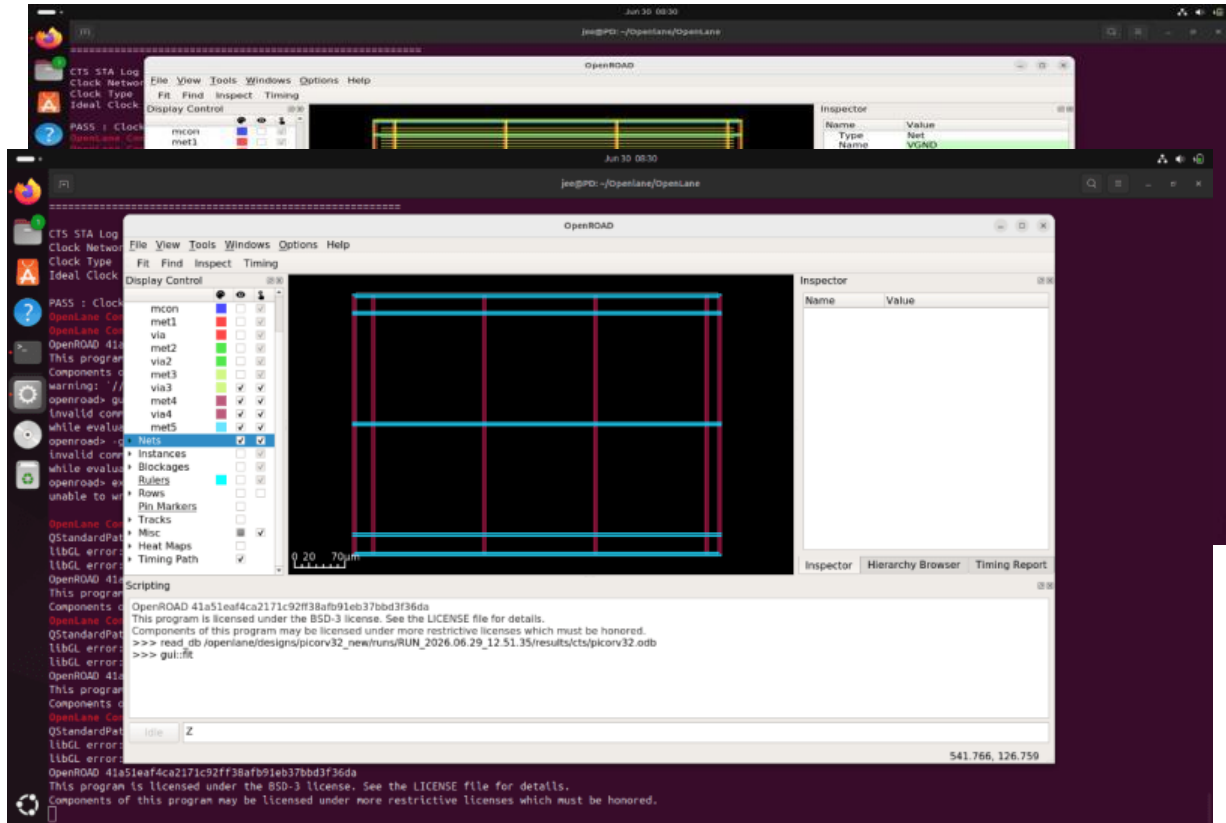
- The PDN consists of horizontal and vertical metal straps connected through vias.
- A continuous core ring is present around the design boundary.
- The highlighted **VGND** net confirms proper ground connectivity across the design.
- The generated power network covers the complete core area and is connected to all standard cells.

Result

The OpenROAD GUI successfully displays the generated **Power Distribution Network (PDN)**. The highlighted **VGND** net verifies that the ground network and core power ring have been created successfully, ensuring proper power delivery throughout the design.



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Metal Layers Visualization

Objective

To visualize the routing metal layers used in the physical design through the OpenROAD GUI.

Procedure

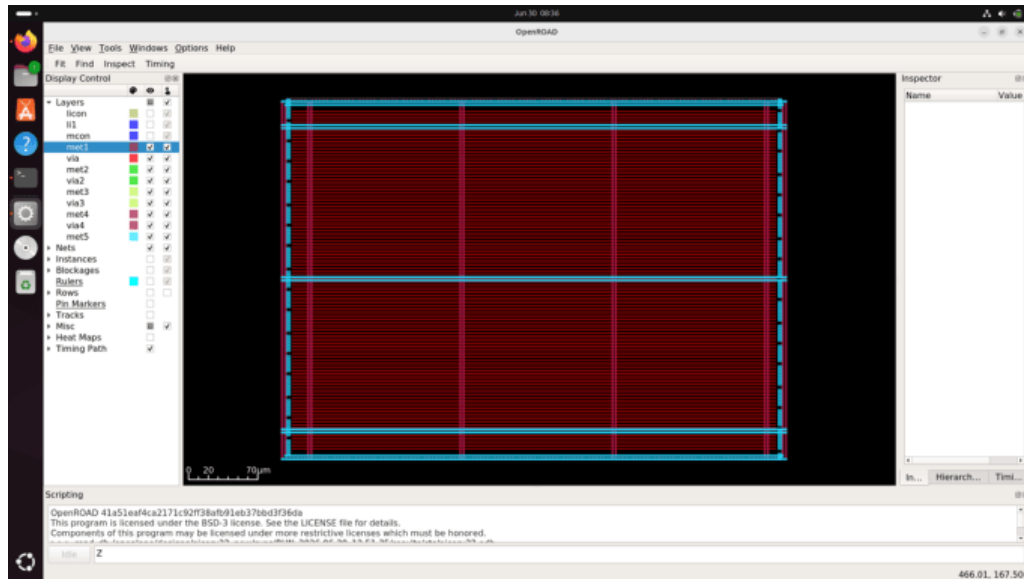
1. Opened the design database (`picorv32.odb`) in the OpenROAD GUI.
2. Enabled the required metal layers (**Metal1 to Metal5**) from the **Display Control** panel.
3. Observed the routing resources and power network across the complete chip layout.

Observation

- **Metal1 (Red):** Used for standard-cell routing inside the core.
- **Metal4 (Magenta):** Used for vertical power straps.
- **Metal5 (Cyan):** Used for horizontal power straps and power ring.
- Multiple metal layers are connected using vias to form the complete routing network.
- The entire chip routing is visible without discontinuities.

Result

The OpenROAD GUI successfully displays the routing across multiple metal layers. The visualization confirms that the routing and power distribution network have been implemented using different metal layers, ensuring proper signal and power connectivity.



Report Different Varieties of Physical-Only Cells

Objective

To identify the physical-only cells inserted during the floorplan stage using the OpenROAD GUI.

Procedure

1. Loaded the floorplan database (`picorv32.odg`) in the OpenROAD GUI.
2. Opened the **Hierarchy Browser**.
3. Expanded the **Physical only** category to observe the inserted physical-only cells.

Observation

The following physical-only cells were identified:

Physical-only Cell	Count
Tap Cells	2349
Fill Cells	250
Total Physical-only Cells	2599



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These cells are inserted automatically during the floorplanning stage to improve manufacturability and power integrity. Tap cells provide substrate connections, while filler cells remove spacing gaps between standard cells.

Result

The OpenROAD floorplan successfully inserted **2599 physical-only cells**, consisting of **2349 tap cells** and **250 filler cells**, confirming proper floorplan preparation.

